电动力学

池灵泽

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目录

符号表 **2**

1 矢量分析 **3**

1.1 定义 3
1.2 微分运算 9
1.2.1 和规则 9
1.2.2 积规则 9
1.2.3 二阶积规则 10
1.3 积分运算 10
1.3.1 高斯定理 10
1.3.2 斯托克斯定理 11
1.4 曲面坐标系 11
1.4.1 柱坐标系 11
1.4.2 球坐标系 12
1.5 狄拉克函数与阶跃函数 13
1.5.1 狄拉克函数 13
1.5.2 阶跃函数 14
1.6 阶跃函数的导数 14
1.6.1 多维狄拉克函数 14
1.7 矢量场理论 14
1.7.1 亥姆霍兹定理 14
1.7.2 势函数 16
1.8 习题 17

2 静电学 **25**

2.1 电场 25
2.2 电势 27
2.2.1 定义 27
2.3 静电场的能量 29
2.4 导体 31
2.5 电容 33
2.6 习题 34

符号表

符号	含义
δ_{ij}	Kronecker 符号
$\cdot$	点乘
$\times$	叉乘
$\boldsymbol{a}$	矢量函数
f	标量函数
a_i	矢量 $\boldsymbol{a}$ 的第 i 个分量
δ_{ij}	克罗内克尔符号
$\delta(\boldsymbol{x})$	狄拉克函数
ε_{ijk}	Levi–Civita 符号 (三维)
ε_0	真空介电常数
x_i	第 i 个坐标分量
$\boldsymbol{e}$	单位矢量
∇	哈密顿算子
r	$\sqrt{x_i x_i}$
$\boldsymbol{r}$	$x_i \boldsymbol{e}_i$
$\boldsymbol{R}$	旋转矩阵
$\boldsymbol{\Phi}_E$	电场强度通量
ρ	电荷密度

1 矢量分析

1.1 定义

1.1.1 爱因斯坦求和约定

$$\sum_{i=1}^{3} a_i b_i \equiv a_i b_i \tag{1.1}$$

1.1.2 克罗内克尔符号

$$\delta_{ij} \equiv \begin{cases} 1 \ (i = j) \\ 0 \ (i \neq j) \end{cases} \tag{1.2}$$

性质:

$$a_i b_j \delta_{ij} = a_i b_i \tag{1.3}$$

$$\delta_{ij} \delta_{jk} = \delta_{ik} \tag{1.4}$$

1.1.3 点乘

$$\boldsymbol{a} \cdot \boldsymbol{b} \equiv a_i b_i \tag{1.5}$$

1.1.4 Levi-Civita 符号

$$\varepsilon_{ijk} \equiv \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix} \tag{1.6}$$

性质:

$$\varepsilon_{ijk} = -\varepsilon_{ikj} = -\varepsilon_{kji} = -\varepsilon_{jik} \tag{1.7}$$

$$
\begin{aligned}
\varepsilon_{ijk}\varepsilon_{lmn} &= \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix} \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix} \\
&= \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix}^T \begin{pmatrix} \delta_{1l} & \delta_{2l} & \delta_{3l} \\ \delta_{1m} & \delta_{2m} & \delta_{3m} \\ \delta_{1n} & \delta_{2n} & \delta_{3n} \end{pmatrix} \\
&= \begin{pmatrix} \delta_{1i} & \delta_{1j} & \delta_{1k} \\ \delta_{2i} & \delta_{2j} & \delta_{2k} \\ \delta_{3i} & \delta_{3j} & \delta_{3k} \end{pmatrix} \begin{pmatrix} \delta_{1l} & \delta_{2l} & \delta_{3l} \\ \delta_{1m} & \delta_{2m} & \delta_{3m} \\ \delta_{1n} & \delta_{2n} & \delta_{3n} \end{pmatrix} \\
&= \begin{pmatrix} \delta_{1l} & \delta_{2l} & \delta_{3l} \\ \delta_{1m} & \delta_{2m} & \delta_{3m} \\ \delta_{1n} & \delta_{2n} & \delta_{3n} \end{pmatrix} \begin{pmatrix} \delta_{1i} & \delta_{1j} & \delta_{1k} \\ \delta_{2i} & \delta_{2j} & \delta_{2k} \\ \delta_{3i} & \delta_{3j} & \delta_{3k} \end{pmatrix} \\
&= \begin{pmatrix} \delta_{li} & \delta_{lj} & \delta_{lk} \\ \delta_{mi} & \delta_{mj} & \delta_{mk} \\ \delta_{ni} & \delta_{nj} & \delta_{nk} \end{pmatrix}
\end{aligned} \tag{1.8}
$$

$$
\varepsilon_{ijk}\varepsilon_{lmk} = \sum_{k=1}^{3} \begin{pmatrix} \delta_{li} & \delta_{lj} & \delta_{lk} \\ \delta_{mi} & \delta_{mj} & \delta_{mk} \\ \delta_{ki} & \delta_{kj} & 1 \end{pmatrix} = \begin{pmatrix} \delta_{li} & \delta_{lj} & 0 \\ \delta_{mi} & \delta_{mj} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \delta_{li}\delta_{mj} - \delta_{lj}\delta_{mi} \tag{1.9}
$$

$$
\varepsilon_{ijk}\varepsilon_{ljk} = \delta_{li}\delta_{jj} - \delta_{lj}\delta_{ji} = 3\delta_{li} - \delta_{li} = 2\delta_{li} \tag{1.10}
$$

1.1.5 叉乘

$$
\boldsymbol{a} \times \boldsymbol{b} \equiv \varepsilon_{ijk} a_i b_j \boldsymbol{e}_k = \begin{pmatrix} \boldsymbol{e}_i & \boldsymbol{e}_j & \boldsymbol{e}_k \\ a_i & a_j & a_k \\ b_i & b_j & b_k \end{pmatrix} \tag{1.11}
$$

性质:

$$
\begin{aligned}
\boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b}) &= \boldsymbol{c}\cdot\varepsilon_{ijk}a_ib_j\boldsymbol{e}_k \\
&= c_k\varepsilon_{ijk}a_ib_j \\
&= \varepsilon_{ijk}a_ib_jc_k \\
&= \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix} a_ib_jc_k \\
&= \begin{pmatrix} \delta_{1i}a_i & \delta_{2i}a_i & \delta_{3i}a_i \\ \delta_{1j}b_j & \delta_{2j}b_j & \delta_{3j}b_j \\ \delta_{1k}c_k & \delta_{2k}c_k & \delta_{3k}c_k \end{pmatrix} \\
&= \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}
\end{aligned} \tag{1.12}
$$

$$
\begin{aligned}
\boldsymbol{c}\times(\boldsymbol{a}\times\boldsymbol{b}) &= \boldsymbol{c}\times\varepsilon_{ijk}a_ib_j\boldsymbol{e}_k \\
&= \varepsilon_{lkn}c_l\varepsilon_{ijk}a_ib_j\boldsymbol{e}_n \\
&= -\varepsilon_{lnk}\varepsilon_{ijk}a_ib_jc_l\boldsymbol{e}_n \\
&= -(\delta_{il}\delta_{jn}-\delta_{in}\delta_{jl})a_ib_jc_l\boldsymbol{e}_n \\
&= -\delta_{il}\delta_{jn}a_ib_jc_l\boldsymbol{e}_n+\delta_{in}\delta_{jl}a_ib_jc_l\boldsymbol{e}_n \\
&= -a_ib_jc_i\boldsymbol{e}_j+a_ib_jc_j\boldsymbol{e}_i \\
&= -(\boldsymbol{a}\cdot\boldsymbol{c})\boldsymbol{b}+(\boldsymbol{b}\cdot\boldsymbol{c})\boldsymbol{a}
\end{aligned} \tag{1.13}
$$

$$
\boldsymbol{e}\times(\boldsymbol{e}\times\boldsymbol{b}) = -(\boldsymbol{e}\cdot\boldsymbol{e})\boldsymbol{b}+(\boldsymbol{b}\cdot\boldsymbol{e})\boldsymbol{e} = -\boldsymbol{b}+(\boldsymbol{b}\cdot\boldsymbol{e})\boldsymbol{e} \tag{1.14}
$$

1.1.6 哈密顿算子

$$
\boldsymbol{\nabla} \equiv \frac{\partial}{\partial x_i}\boldsymbol{e}_i \tag{1.15}
$$

1.1.7 梯度

设位置矢量 $\boldsymbol{r}=\boldsymbol{r}(u_1,u_2,u_3)$, 拉梅系数为

$$
h_i \equiv \left\|\frac{\partial\boldsymbol{r}}{\partial u_i}\right\|, \qquad \boldsymbol{e} \equiv \frac{1}{h_i}\frac{\partial\boldsymbol{r}}{\partial u_i}
$$

且 $\boldsymbol{e}_i\cdot\boldsymbol{e}_j=\delta_{ij}$。

对标量场 $f(u_1,u_2,u_3)$, 全微分为

$$
\mathrm{d}f = \frac{\partial f}{\partial u_i}\,\mathrm{d}u_i.
$$

无穷小位移 (线元) 为

$$\mathrm{d}\boldsymbol{l} = h_i\ \mathrm{d}u_i\,\boldsymbol{e}_i.$$

设 $\boldsymbol{\nabla} f = G_i\boldsymbol{e}_i$, 则

$$\mathrm{d}f = \boldsymbol{\nabla} f\cdot\boldsymbol{dl} = G_i h_i\ \mathrm{d}u_i.$$

比较系数, 得 $\boldsymbol{G} = \dfrac{1}{h_i}\dfrac{\partial f}{\partial u_i}\boldsymbol{e}_i$, 从而

$$\boldsymbol{\nabla} f = \frac{1}{h_i}\frac{\partial f}{\partial u_i}\boldsymbol{e}_i \tag{1.16}$$

对于直角坐标

$$\boldsymbol{\nabla} f = \frac{\partial f}{\partial x_i}\boldsymbol{e}_i \tag{1.17}$$

性质:

$$\boldsymbol{\nabla} r = \frac{\partial\sqrt{x_i x_i}}{\partial x_i}\boldsymbol{e}_i = \frac{x_i}{r}\boldsymbol{e}_i = \frac{\boldsymbol{r}}{r} \tag{1.18}$$

$$\boldsymbol{\nabla} f(r) = \frac{\partial f}{\partial r}\frac{\partial r}{\partial x_i}\boldsymbol{e}_i = \frac{\partial f}{\partial r}\frac{\boldsymbol{r}}{r} \tag{1.19}$$

1.1.8 散度

对微小增量 $\mathrm{d}u_1, \mathrm{d}u_2, \mathrm{d}u_3$, 线元在三方向的实际长度分别为

$$\mathrm{d}l_1 = h_1\,\mathrm{d}u_1,\quad \mathrm{d}l_2 = h_2\,\mathrm{d}u_2,\quad \mathrm{d}l_3 = h_3\,\mathrm{d}u_3$$

因此微小长方体的体元为

$$\mathrm{d}\tau = \mathrm{d}l_1\,\mathrm{d}l_2\,\mathrm{d}l_3 = h_1h_2h_3\,\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3. \tag{1.20}$$

设矢量场在该点为

$$\boldsymbol{a} = a_1\boldsymbol{e}_1 + a_2\boldsymbol{e}_2 + a_3\boldsymbol{e}_3,$$

其中 a_1, a_2, a_3 为在局部单位基矢下的分量 (均为 u_1, u_2, u_3 的函数)。

下面计算穿过微小长方体六个面的净通量。以 u_1 方向为例, 考察在 u_1 与 $u_1 + \mathrm{d}u_1$ 两个面:

面 u_1(在坐标 u 处, 向外法向为 $-\boldsymbol{e}_1$)：面积元为

$$\mathrm{d}\boldsymbol{a}_1^{(-)} = -\boldsymbol{e}_1(h_2\,\mathrm{d}u_2)(h_3\,\mathrm{d}u_3) = -\boldsymbol{e}_1h_2h_3\,\mathrm{d}u_2\,\mathrm{d}u_3,$$

对应的通量 (近似取该面中心处分量)：

$$\Phi_1^{(-)} = \boldsymbol{a}(u_1,u_2,u_3)\cdot\mathrm{d}\boldsymbol{a}_1^{(-)} = -a_1(u_1,u_2,u_3)h_2h_3\,\mathrm{d}u_2\,\mathrm{d}u_3.$$

面 $u + \mathrm{d}u$(向外法向为 $+\boldsymbol{e}_1$)：面积元

$$\mathrm{d}\boldsymbol{a}_1^{(+)} = +\boldsymbol{e}_1h_2h_3\,\mathrm{d}u_2\,\mathrm{d}u_3,$$

通量 (在 $u + du$ 处)：

$$\Phi_1^{(+)} = a_1(u_1+du_1,u_2,u_3)h_2(u_1+du_1,u_2,u_3)h_3(u_1+du_1,u_2,u_3)\,\mathrm{d}u_2\,\mathrm{d}u_3.$$

因此穿过这对面的净通量为

$$\begin{aligned}\Delta\Phi_1 &= \Phi_1^{(+)} - \Phi_1^{(-)}\\ &= [a_1(u_1+\mathrm{d}u_1,u_2,u_3)h_2(u_1+\mathrm{d}u_1,u_2,u_3)h_3(u_1+\mathrm{d}u_1,u_2,u_3) - a_1(u_1,u_2,u_3)h_2h_3]\,\mathrm{d}u_2\,\mathrm{d}u_3.\end{aligned}$$

用泰勒展开到一阶并忽略高阶项, 得

$$\Delta\Phi_u = \frac{\partial}{\partial u_1}(h_2h_3a_1)\,\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3 + o(\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3).$$

对 u_2 和 u_3 方向做同样的计算, 六个面的净通量近似为三项之和：

$$\Delta\Phi_{\text{total}} = \frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{a_i}{h_i}\right)\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3$$

由于体元按照式(1.20)为 $\mathrm{d}\tau = h_1h_2h_3\,\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3$, 故单位体积的通量密度 (即散度) 为极限：

$$\begin{aligned}\boldsymbol{\nabla}\cdot\boldsymbol{a} &= \lim_{\Delta\tau\to 0}\frac{\Delta\Phi_{\text{total}}}{d\tau}\\ &= \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{a_i}{h_i}\right).\end{aligned}$$

于是得到正交曲线坐标系下的散度公式：

$$\boldsymbol{\nabla}\cdot\boldsymbol{a} = \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{a_i}{h_i}\right). \tag{1.21}$$

对于直角坐标

$$\boldsymbol{\nabla}\cdot\boldsymbol{a} = \frac{\partial a_i}{\partial x_i} \tag{1.22}$$

性质:

$$\boldsymbol{\nabla}\cdot\boldsymbol{r} = 3 \tag{1.23}$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{f}(r) = \frac{\partial f_i(r)}{\partial x_i} = f_i'\frac{\partial r}{\partial x_i} = \boldsymbol{f}'\cdot\frac{\boldsymbol{r}}{r} \tag{1.24}$$

1.1.9 旋度

选取微小矩形位于 u_1-u_2 平面, 固定 u_3。矩形四边的线元素：

$$\begin{aligned}
&\text{下边: } \mathrm{d}\boldsymbol{l}_1 = h_1\,\mathrm{d}u_1\boldsymbol{e}_1, \quad \text{从}(u_1,u_2)\to(u_1+\mathrm{d}u_1,u_2)\\
&\text{右边: } \mathrm{d}\boldsymbol{l}_2 = h_2\,\mathrm{d}u_2\boldsymbol{e}_2, \quad \text{从}(u_1+\mathrm{d}u_1,u_2)\to(u_1+\mathrm{d}u_1,u_2+\mathrm{d}u_2)\\
&\text{上边: } \mathrm{d}\boldsymbol{l}_3 = -h_1\,\mathrm{d}u_1\boldsymbol{e}_1, \quad \text{从}(u_1+\mathrm{d}u_1,u_2+\mathrm{d}u_2)\to(u_1,u_2+\mathrm{d}u_2)\\
&\text{左边: } \mathrm{d}\boldsymbol{l}_4 = -h_2du_2\boldsymbol{e}_2, \quad \text{从}(u_1,u_2+\mathrm{d}u_2)\to(u_1,u_2)
\end{aligned}$$

沿每一边计算 $\boldsymbol{a}\cdot d\boldsymbol{l}$ 并做泰勒展开 (只保留一阶项)：

$$\oint\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l} \approx \left[\frac{\partial}{\partial u_1}(h_2a_2) - \frac{\partial}{\partial u_2}(h_1a_1)\right]\mathrm{d}u_1\,\mathrm{d}u_2$$

面元面积：

$$dS_3 = h_1h_2\,\mathrm{d}u_1\,\mathrm{d}u_2$$

旋度在 $\boldsymbol{e}_3$ 方向的分量：

$$(\boldsymbol{\nabla}\times\boldsymbol{a})_3 = \lim_{\mathrm{d}u_1,\mathrm{d}u_2\to 0}\frac{\oint\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l}}{\mathrm{d}S_3} = \frac{1}{h_1h_2}\left[\frac{\partial}{\partial u_1}(h_2a_2) - \frac{\partial}{\partial u_2}(h_1a_1)\right]$$

同理, 对于其他两个方向：

$$(\boldsymbol{\nabla}\times\boldsymbol{a})_1 = \frac{1}{h_2h_3}\left[\frac{\partial}{\partial u_2}(h_3a_3) - \frac{\partial}{\partial u_3}(h_2a_2)\right], \quad (\boldsymbol{\nabla}\times\boldsymbol{a})_2 = \frac{1}{h_3h_1}\left[\frac{\partial}{\partial u_3}(h_1a_1) - \frac{\partial}{\partial u_1}(h_3a_3)\right]$$

最终正交曲线坐标系下旋度公式为：

$$\boldsymbol{\nabla} \times \boldsymbol{a} = \frac{h_i}{h_1 h_2 h_3} \varepsilon_{ijk} \frac{\partial (h_k a_k)}{\partial u_j} \boldsymbol{e}_i \tag{1.25}$$

对于直角坐标

$$\boldsymbol{\nabla} \times \boldsymbol{a} = \varepsilon_{ijk} \frac{\partial a_j}{\partial x_i} \boldsymbol{e}_k \tag{1.26}$$

性质:

$$\boldsymbol{\nabla} \times \boldsymbol{r} = 0 \tag{1.27}$$

$$\boldsymbol{\nabla} \times (f\boldsymbol{a}) = \varepsilon_{ijk} \frac{\partial (f a_j)}{\partial x_i} \boldsymbol{e}_k = f \varepsilon_{ijk} \frac{\partial a_j}{\partial x_i} \boldsymbol{e}_k + \varepsilon_{ijk} \frac{\partial f}{\partial x_i} a_j \boldsymbol{e}_k = f(\boldsymbol{\nabla} \times \boldsymbol{a}) + (\boldsymbol{\nabla} f) \times \boldsymbol{a} \tag{1.28}$$

1.1.10 并积

$$\boldsymbol{\nabla} \boldsymbol{a} \equiv \begin{pmatrix} \frac{\partial a_1}{x_1} & \frac{\partial a_2}{x_1} & \frac{\partial a_3}{x_1} \\ \frac{\partial a_1}{x_2} & \frac{\partial a_2}{x_2} & \frac{\partial a_3}{x_2} \\ \frac{\partial a_1}{x_3} & \frac{\partial a_2}{x_3} & \frac{\partial a_3}{x_3} \end{pmatrix} = \frac{\partial a_i}{\partial x_j} \boldsymbol{e}_i \boldsymbol{e}_j \tag{1.29}$$

1.1.11 拉普拉斯算子

定义为梯度的散度, 由1.16和1.21得

$$\begin{aligned} \boldsymbol{\nabla}^2 f &= \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} f) \\ &= \boldsymbol{\nabla} \cdot \left(\frac{1}{h_i} \frac{\partial f}{\partial u_i} \right) \\ &= \frac{1}{h_1 h_2 h_3} \frac{\partial}{\partial u_i} \left(h_1 h_2 h_3 \frac{\frac{1}{h_i} \frac{\partial f}{\partial u_i}}{h_i} \right) \\ &= \frac{1}{h_1 h_2 h_3} \frac{\partial}{\partial u_i} \left(h_1 h_2 h_3 \frac{\partial f}{h_i^2 \partial u_i} \right) \end{aligned} \tag{1.30}$$

对于直角坐标

$$\boldsymbol{\nabla}^2 = \frac{\partial^2}{\partial x_i^2} \tag{1.31}$$

性质:

$$
\begin{aligned}
\boldsymbol{\nabla}^2(fg) &= \frac{\partial}{\partial x_i}\frac{\partial fg}{\partial x_i} \\
&= \frac{\partial}{\partial x_i}\left(g\frac{\partial f}{\partial x_i} + f\frac{\partial g}{\partial x_i}\right) \\
&= \frac{\partial}{\partial x_i}\left(g\frac{\partial f}{\partial x_i}\right) + \frac{\partial}{\partial x_i}\left(f\frac{\partial g}{\partial x_i}\right) \\
&= \frac{\partial}{\partial x_i}\left(g\frac{\partial f}{\partial x_i}\right) + \frac{\partial}{\partial x_i}\left(f\frac{\partial g}{\partial x_i}\right) \\
&= \frac{\partial g}{\partial x_i}\frac{\partial f}{\partial x_i} + g\frac{\partial^2 f}{\partial x_i^2} + \frac{\partial f}{\partial x_i}\frac{\partial g}{\partial x_i} + a\frac{\partial^2 g}{\partial x_i^2} \\
&= g\frac{\partial^2 f}{\partial x_i^2} + 2\frac{\partial f}{\partial x_i}\frac{\partial g}{\partial x_i} + f\frac{\partial^2 g}{\partial x_i^2} \\
&= g\boldsymbol{\nabla}^2 f + 2(\boldsymbol{\nabla} f)\cdot(\boldsymbol{\nabla} g) + f\boldsymbol{\nabla}^2 g
\end{aligned} \tag{1.32}
$$

1.2 微分运算

1.2.1 和规则

$$\boldsymbol{\nabla}(f+g) = \boldsymbol{\nabla} f + \boldsymbol{\nabla} g \tag{1.33}$$

$$\boldsymbol{\nabla}\cdot(\boldsymbol{a}+\boldsymbol{b}) = \boldsymbol{\nabla}\cdot\boldsymbol{a} + \boldsymbol{\nabla}\cdot\boldsymbol{b} \tag{1.34}$$

$$\boldsymbol{\nabla}\times(\boldsymbol{a}+\boldsymbol{b}) = \boldsymbol{\nabla}\times\boldsymbol{a} + \boldsymbol{\nabla}\times\boldsymbol{b} \tag{1.35}$$

1.2.2 积规则

$$\boldsymbol{\nabla}(fg) = g\boldsymbol{\nabla} f + f\boldsymbol{\nabla} g \tag{1.36}$$

$$
\begin{aligned}
&\boldsymbol{a}\times(\boldsymbol{\nabla}\times\boldsymbol{b}) + \boldsymbol{b}\times(\boldsymbol{\nabla}\times\boldsymbol{a}) + (\boldsymbol{a}\cdot\boldsymbol{\nabla})\boldsymbol{b} + (\boldsymbol{b}\cdot\boldsymbol{\nabla})\boldsymbol{a} \\
=&\boldsymbol{a}\times\left(\varepsilon_{ijk}\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_k\right) + \boldsymbol{b}\times\left(\varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_k\right) + \left(a_i\frac{\partial}{\partial x_i}\right)\boldsymbol{b} + \left(b_i\frac{\partial}{\partial x_i}\right)\boldsymbol{a} \\
=&\varepsilon_{lkn}a_l\varepsilon_{ijk}\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n + \varepsilon_{lkn}b_l\varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&-\varepsilon_{lnk}\varepsilon_{ijk}a_l\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n - \varepsilon_{lnk}\varepsilon_{ijk}b_l\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&-(\delta_{il}\delta_{jn} - \delta_{in}\delta_{jl})a_l\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n - (\delta_{il}\delta_{jn} - \delta_{in}\delta_{jl})b_l\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&\left(-\delta_{il}\delta_{jn}a_l\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n + \delta_{in}\delta_{jl}a_l\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n\right) + \left(-\delta_{il}\delta_{jn}b_l\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n + \delta_{in}\delta_{jl}b_l\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n\right) + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&\left(-a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + a_j\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_i\right) + \left(-b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j + b_j\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_i\right) + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&a_j\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_i + b_j\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_i \\
=&\frac{\partial(a_j b_j)}{\partial x_i}\boldsymbol{e}_i \\
=&\boldsymbol{\nabla}(\boldsymbol{a}\cdot\boldsymbol{b})
\end{aligned} \tag{1.37}
$$

$$\boldsymbol{\nabla}\cdot(f\boldsymbol{a})=\frac{\partial f a_i}{\partial x_i}=f\frac{\partial a_i}{\partial x_i}+a_i\frac{\partial f}{\partial x_i}=f\boldsymbol{\nabla}\cdot\boldsymbol{a}+\boldsymbol{a}\cdot\boldsymbol{\nabla}f \tag{1.38}$$

$$\begin{aligned}\boldsymbol{b}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})-\boldsymbol{a}\cdot(\boldsymbol{\nabla}\times\boldsymbol{b})&=\boldsymbol{b}\cdot(\varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_k)-\boldsymbol{a}\cdot(\varepsilon_{ijk}\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_k)\\&=b_k\varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}-a_k\varepsilon_{ijk}\frac{\partial b_j}{\partial x_i}\\&=b_j\varepsilon_{ijk}\frac{\partial a_i}{\partial x_k}+a_i\varepsilon_{ijk}\frac{\partial b_j}{\partial x_k}\\&=\frac{\partial\varepsilon_{ijk}a_ib_j}{\partial x_k}\\&=\boldsymbol{\nabla}\cdot(\varepsilon_{ijk}a_ib_j\boldsymbol{e}_k)\\&=\boldsymbol{\nabla}\cdot(\boldsymbol{a}\times\boldsymbol{b})\end{aligned} \tag{1.39}$$

1.2.3 二阶积规则

$$\boldsymbol{\nabla}\times(\boldsymbol{\nabla}f)=\boldsymbol{\nabla}\times(\frac{\partial f}{\partial x_i}\boldsymbol{e}_i)=\varepsilon_{ijk}\frac{\partial}{\partial x_i}\frac{\partial f}{\partial x_j}\boldsymbol{e}_k=0 \tag{1.40}$$

$$\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})=\boldsymbol{\nabla}\cdot(\varepsilon_{ijk}\frac{\partial a_i}{\partial x_j}\boldsymbol{e}_k)=\varepsilon_{ijk}\frac{\partial^2 a_i}{\partial x_j\partial x_k}=0 \tag{1.41}$$

$$\Delta f=\frac{\partial^2 f}{\partial x_i^2} \tag{1.42}$$

$$\Delta\boldsymbol{a}=\frac{\partial^2(a_j\boldsymbol{e}_j)}{\partial x_i^2} \tag{1.43}$$

$$\begin{aligned}\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{a})&=\boldsymbol{\nabla}\times(\varepsilon_{ijk}\frac{\partial a_i}{\partial x_j}\boldsymbol{e}_k)\\&=\varepsilon_{lmn}\varepsilon_{ijl}\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n\\&=\varepsilon_{mnl}\varepsilon_{ijl}\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n\\&=(\delta_{im}\delta_{jn}-\delta_{in}\delta_{jm})\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n\\&=\delta_{im}\delta_{jn}\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n-\delta_{in}\delta_{jm}\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n\\&=\frac{\partial^2 a_i}{\partial x_j\partial x_i}\boldsymbol{e}_j-\frac{\partial^2 a_i}{\partial x_j\partial x_j}\boldsymbol{e}_i\\&=\boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{a})-\Delta\boldsymbol{a}\end{aligned} \tag{1.44}$$

1.3 积分运算

1.3.1 高斯定理

我们已经得到正交坐标系中的散度局部公式1.21。因此, 对体积元 $\Delta\tau=h_1h_2h_3\,\Delta u_1\Delta u_2\Delta u_3$ 的六个面计算净通量时：

$$\Phi^{(\mathrm{out})}-\Phi^{(\mathrm{in})}=\left[\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{a_i}{h_i}\right)\right]\Delta u_1\Delta u_2\Delta u_3=(\boldsymbol{\nabla}\cdot\boldsymbol{a})\,\Delta\tau.$$

将整个区域 V 分割成许多此类体积元, 内部公共面的通量相互抵消, 仅留下外边界 ∂V 的通量。令分割尺寸趋于零, 通量和体积和分别趋于各自积分, 故得到

$$\iiint_V (\boldsymbol{\nabla}\cdot\boldsymbol{a})\,\mathrm{d}V = \iint_{\partial V} \boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S} \tag{1.45}$$

1.3.2 斯托克斯定理

我们已经得到任意正交坐标系中的旋度局部公式1.25 , 考虑由 (u_1, u_2) 增量生成的一个无穷小曲面元

$$\mathrm{d}\boldsymbol{S} = (h_1 h_2\,\mathrm{d}u_1\,\mathrm{d}u_2)\boldsymbol{e}_3.$$

沿此小曲面元的边界进行线积分, 可得

$$\oint\!\!\!\oint_{\partial S} \boldsymbol{a}\cdot d\boldsymbol{l} = (\nabla\times\boldsymbol{a})\cdot d\boldsymbol{S}.$$

将整个曲面分割成许多这样的小元, 内部公共边界的线积分相互抵消, 仅留下外边界 ∂S 的线积分。令分割尺寸趋于零, 总和趋于积分, 故得

$$\iint_S (\boldsymbol{\nabla}\times\boldsymbol{a})\cdot\mathrm{d}\boldsymbol{S} = \oint\!\!\!\oint_{\partial S} \boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l} \tag{1.46}$$

1.4 曲面坐标系

1.4.1 柱坐标系

将柱坐标系坐标转化为直角系坐标的方式为

$$x_1 = s\cos\phi,\ x_2 = s\sin\phi,\ x_3 = x_3 \tag{1.47}$$

其中 s 是坐标点到 z 轴的距离。柱坐标系和直角系坐标无限小元转换方式为

$$\begin{pmatrix}\mathrm{d}x_1\\ \mathrm{d}x_2\\ \mathrm{d}x_3\end{pmatrix} = \begin{pmatrix}\frac{x_1}{s} & -x_2 & 0\\ \frac{x_2}{s} & x_1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}\mathrm{d}s\\ \mathrm{d}\phi\\ \mathrm{d}x_3\end{pmatrix} \tag{1.48}$$

其拉梅系数为

$$h_1 = 1,\ h_2 = s,\ h_3 = 1 \tag{1.49}$$

其无限小位移为

$$\mathrm{d}\boldsymbol{l} = h_i\,\mathrm{d}u_i\boldsymbol{e}_i = \mathrm{d}s\boldsymbol{s} + s\,\mathrm{d}\phi\boldsymbol{\phi} + \mathrm{d}x_3\boldsymbol{x}_3 \tag{1.50}$$

其体积元为

$$\mathrm{d}V = s\,\mathrm{d}s\,\mathrm{d}\phi\,\mathrm{d}x_3 \tag{1.51}$$

梯度

$$\nabla f = \frac{1}{h_i}\frac{\partial f}{\partial u_i}\boldsymbol{e}_i = \frac{\partial}{\partial s}\boldsymbol{s} + \frac{1}{s}\frac{\partial}{\partial\phi}\boldsymbol{\phi} + \frac{\partial}{\partial x_3}\boldsymbol{x}_3 \tag{1.52}$$

散度

$$\boldsymbol{\nabla}\cdot\boldsymbol{a} = \frac{1}{s}(\frac{\partial s a_s}{\partial s} + \frac{\partial a_\phi}{\partial\phi} + \frac{\partial s a_3}{\partial x_3}) = \frac{1}{s}\frac{\partial s a_s}{\partial s} + \frac{1}{s}\frac{\partial a_\phi}{\partial\phi} + \frac{\partial a_3}{\partial x_3} = \frac{a_s}{s} + \frac{\partial a_s}{\partial s} + \frac{1}{s}\frac{\partial a_\phi}{\partial\phi} + \frac{\partial a_3}{\partial x_3} \tag{1.53}$$

旋度

$$
\begin{aligned}
\boldsymbol{\nabla}\times\boldsymbol{a} &= \frac{h_i}{h_1h_2h_3}\varepsilon_{ijk}\frac{\partial(h_ka_k)}{\partial u_j}\boldsymbol{e}_i \\
&= \frac{h_i}{s}\varepsilon_{ijk}\frac{\partial(h_ka_k)}{\partial u_j}\boldsymbol{e}_i \\
&= \frac{1}{s}\left(\varepsilon_{123}\frac{\partial a_3}{\partial u_2}\boldsymbol{e}_1+\varepsilon_{132}\frac{\partial sa_2}{\partial u_3}\boldsymbol{e}_1+\varepsilon_{321}\frac{\partial a_1}{\partial u_2}\boldsymbol{e}_3+s\varepsilon_{213}\frac{\partial a_3}{\partial u_1}\boldsymbol{e}_2+\varepsilon_{312}\frac{\partial sa_2}{\partial u_1}\boldsymbol{e}_3+s\varepsilon_{231}\frac{\partial a_1}{\partial u_3}\boldsymbol{e}_2\right) \\
&= \frac{1}{s}\left(\frac{\partial a_3}{\partial u_2}\boldsymbol{e}_1-\frac{\partial sa_2}{\partial u_3}\boldsymbol{e}_1-\frac{\partial a_1}{\partial u_2}\boldsymbol{e}_3-s\frac{\partial a_3}{\partial u_1}\boldsymbol{e}_2+\frac{\partial sa_2}{\partial u_1}\boldsymbol{e}_3+s\frac{\partial a_1}{\partial u_3}\boldsymbol{e}_2\right) \\
&= \frac{1}{s}\left(\frac{\partial a_3}{\partial\phi}\boldsymbol{s}-\frac{\partial sa_\phi}{\partial x_3}\boldsymbol{s}-\frac{\partial a_s}{\partial\phi}\boldsymbol{e}_3-s\frac{\partial a_3}{\partial s}\boldsymbol{\phi}+\frac{\partial sa_\phi}{\partial s}\boldsymbol{e}_3+s\frac{\partial a_s}{\partial x_3}\boldsymbol{\phi}\right) \\
&= \frac{1}{s}\left(\frac{\partial a_3}{\partial\phi}-s\frac{\partial a_\phi}{\partial x_3}\right)\boldsymbol{s}+\left(\frac{\partial a_s}{\partial x_3}-\frac{\partial a_3}{\partial s}\right)\boldsymbol{\phi}+\frac{1}{s}\left(\frac{\partial sa_\phi}{\partial s}-\frac{\partial a_s}{\partial\phi}\right)\boldsymbol{e}_3
\end{aligned}
\tag{1.54}
$$

拉普拉斯算子

$$
\begin{aligned}
\boldsymbol{\nabla}^2 f &= \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{\partial f}{h_i^2\partial u_i}\right) \\
&= \frac{1}{s}\frac{\partial}{\partial u_i}\left(s\frac{\partial f}{h_i^2\partial u_i}\right) \\
&= \frac{1}{s}\frac{\partial}{\partial u_s}(s\frac{\partial f}{h_s^2\partial u_s})+\frac{1}{s}\frac{\partial}{\partial\phi}(s\frac{\partial f}{h_\phi^2\partial\phi})+\frac{1}{s}\frac{\partial}{\partial u_3}(s\frac{\partial f}{h_3^2\partial u_3}) \\
&= \frac{1}{s}\frac{\partial}{\partial u_s}(s\frac{\partial f}{\partial u_s})+\frac{1}{s}\frac{\partial}{\partial\phi}(s\frac{\partial f}{s^2\partial\phi})+\frac{1}{s}\frac{\partial}{\partial u_3}(s\frac{\partial f}{\partial u_3}) \\
&= \frac{1}{s}\frac{\partial}{\partial u_s}(s\frac{\partial f}{\partial u_s})+\frac{1}{s^2}\frac{\partial^2 f}{\partial\phi^2}+\frac{\partial^2 f}{\partial u_3^2}
\end{aligned}
\tag{1.55}
$$

1.4.2 球坐标系

将球坐标系坐标转化为直角系坐标的方式为

$$
x_1 = r\cos\phi\sin\theta,\ x_2 = r\sin\phi\sin\theta,\ x_3 = r\cos\theta \tag{1.56}
$$

球坐标系和直角系坐标无限小元转换方式为

$$
\begin{pmatrix}\mathrm{d}x_1\\ \mathrm{d}x_2\\ \mathrm{d}x_3\end{pmatrix}=\begin{pmatrix}\frac{x_1}{r} & -x_2 & \frac{x_1x_3}{r\sin\theta}\\ \frac{x_2}{r} & x_1 & \frac{x_2x_3}{r\sin\theta}\\ \frac{x_3}{r} & 0 & -r\sin\theta\end{pmatrix}\begin{pmatrix}\mathrm{d}r\\ \mathrm{d}\phi\\ \mathrm{d}\theta\end{pmatrix} \tag{1.57}
$$

其拉梅系数为

$$
h_1 = 1,\ h_2 = r\sin\theta,\ h_3 = r \tag{1.58}
$$

其无限小位移为

$$
\mathrm{d}\boldsymbol{l} = \mathrm{d}r\boldsymbol{r} + r\sin\theta\,\mathrm{d}\phi\boldsymbol{\phi} + r\,\mathrm{d}\theta\boldsymbol{\theta} \tag{1.59}
$$

其体积元为

$$
\mathrm{d}V = r\sin\theta r\,\mathrm{d}r\,\mathrm{d}\phi\,\mathrm{d}\theta \tag{1.60}
$$

梯度

$$
\nabla f = \frac{1}{h_i}\frac{\partial f}{\partial u_i}\boldsymbol{e}_i = \frac{\partial}{\partial r}\boldsymbol{r} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}\boldsymbol{\phi} + \frac{1}{r}\frac{\partial}{\partial\theta}\boldsymbol{\theta} \tag{1.61}
$$

散度

$$
\begin{aligned}
\boldsymbol{\nabla}\cdot\boldsymbol{a} &= \frac{1}{r^2\sin\theta}\left(\frac{\partial r^2\sin\theta a_r}{\partial r}+\frac{\partial r a_\phi}{\partial\phi}+\frac{\partial r\sin\theta a_\theta}{\partial\theta}\right)\\
&= \frac{1}{r^2\sin\theta}\left(\sin\theta\frac{\partial r^2 a_r}{\partial r}+r\frac{\partial a_\phi}{\partial\phi}+r\frac{\partial\sin\theta a_\theta}{\partial\theta}\right)\\
&= \frac{1}{r^2}\frac{\partial r^2 a_r}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial a_\phi}{\partial\phi}+\frac{1}{r\sin\theta}\frac{\partial\sin\theta a_\theta}{\partial\theta}
\end{aligned}
\tag{1.62}
$$

旋度

$$
\begin{aligned}
&\boldsymbol{\nabla}\times\boldsymbol{a} \qquad (1.63)\\
=&\frac{h_i}{h_1h_2h_3}\varepsilon_{ijk}\frac{\partial(h_ka_k)}{\partial u_j}\boldsymbol{e}_i\\
=&\frac{h_i}{r^2\sin\theta}\varepsilon_{ijk}\frac{\partial(h_ka_k)}{\partial u_j}\boldsymbol{e}_i\\
=&\frac{1}{r^2\sin\theta}\left(\varepsilon_{123}\frac{\partial ra_3}{\partial u_2}\boldsymbol{e}_1+\varepsilon_{132}\frac{\partial sa_2}{\partial u_3}\boldsymbol{e}_1+r\varepsilon_{321}\frac{\partial a_1}{\partial u_2}\boldsymbol{e}_3+s\varepsilon_{213}\frac{\partial ra_3}{\partial u_1}\boldsymbol{e}_2+r\varepsilon_{312}\frac{\partial sa_2}{\partial u_1}\boldsymbol{e}_3+s\varepsilon_{231}\frac{\partial a_1}{\partial u_3}\boldsymbol{e}_2\right)\\
=&\frac{1}{r^2\sin\theta}\left(\frac{\partial ra_3}{\partial u_2}\boldsymbol{e}_1-\frac{\partial r\sin\theta a_2}{\partial u_3}\boldsymbol{e}_1-r\frac{\partial a_1}{\partial u_2}\boldsymbol{e}_3-r\sin\theta\frac{\partial ra_3}{\partial u_1}\boldsymbol{e}_2+r\frac{\partial r\sin\theta a_2}{\partial u_1}\boldsymbol{e}_3+r\sin\theta\frac{\partial a_1}{\partial u_3}\boldsymbol{e}_2\right)\\
=&\frac{1}{r^2\sin\theta}\left(\frac{\partial ra_\theta}{\partial\phi}\boldsymbol{r}-\frac{\partial r\sin\theta a_\phi}{\partial\theta}\boldsymbol{r}-r\frac{\partial a_r}{\partial\phi}\boldsymbol{\theta}-r\sin\theta\frac{\partial ra_\theta}{\partial r}\boldsymbol{\phi}+r\frac{\partial r\sin\theta a_\phi}{\partial r}\boldsymbol{\theta}+r\sin\theta\frac{\partial a_r}{\partial\theta}\boldsymbol{\phi}\right)\\
=&\frac{1}{r^2\sin\theta}\left(r\frac{\partial a_\theta}{\partial\phi}-r\frac{\partial\sin\theta a_\phi}{\partial\theta}\right)\boldsymbol{r}+\frac{1}{r}\left(\frac{\partial a_r}{\partial\theta}-\frac{\partial ra_\theta}{\partial r}\right)\boldsymbol{\phi}+\frac{1}{r\sin\theta}\left(\sin\theta\frac{\partial ra_\phi}{\partial r}-\frac{\partial a_r}{\partial\phi}\right)\boldsymbol{\theta}\\
=&\frac{1}{r\sin\theta}\left(\frac{\partial a_\theta}{\partial\phi}-\frac{\partial\sin\theta a_\phi}{\partial\theta}\right)\boldsymbol{r}+\frac{1}{r}\left(\frac{\partial a_r}{\partial\theta}-\frac{\partial ra_\theta}{\partial r}\right)\boldsymbol{\phi}+\frac{1}{r\sin\theta}\left(\sin\theta\frac{\partial ra_\phi}{\partial r}-\frac{\partial a_r}{\partial\phi}\right)\boldsymbol{\theta} \qquad (1.64)
\end{aligned}
$$

拉普拉斯算子

$$
\begin{aligned}
\boldsymbol{\nabla}^2 f &= \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{\partial f}{h_i^2\partial u_i}\right)\\
&= \frac{1}{r^2\sin\theta}\frac{\partial}{\partial u_i}\left(r^2\sin\theta\frac{\partial f}{h_i^2\partial u_i}\right)\\
&= \frac{1}{r^2\sin\theta}\frac{\partial}{\partial r}\left(r^2\sin\theta\frac{\partial f}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\phi}\left(\frac{\partial f}{\sin\theta\partial\phi}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right)\\
&= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\phi^2}+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right)
\end{aligned}
\tag{1.65}
$$

1.5 狄拉克函数与阶跃函数

1.5.1 狄拉克函数

严格来说, 狄拉克 δ 函数并不是一个通常意义下的函数, 而是通过其在积分中的作用来定义的。设 $f(x)$ 是在 $x=0$ 附近连续, 并且在无穷远处足够快衰减的函数。定义狄拉克 δ 满足

$$
\int_{-\infty}^{\infty}\delta(x)\,f(x)\,\mathrm{d}x=f(0), \tag{1.66}
$$

由定义立即得到

$$
\int_{-\infty}^{\infty}\delta(x-a)\,f(x)\,\mathrm{d}x=f(a). \tag{1.67}
$$

设 $g(x)$ 在 x_i 处有孤立零点, 且 $g'(x_i)\neq 0$, 则

$$
\delta(g(x))=\sum_i\frac{\delta(x-x_i)}{|g'(x_i)|}. \tag{1.68}
$$

1.5.2 阶跃函数

定义阶跃函数

$$H(x)=\begin{cases}0, & x<0,\\ 1, & x>0.\end{cases} \tag{1.69}$$

1.6 阶跃函数的导数

在通常意义下, $H(x)$ 在 $x=0$ 不可导。但在积分意义下, 我们定义

$$\int_{-\infty}^{\infty}\frac{\mathrm{d}H(x)}{\mathrm{d}x}f(x)\,\mathrm{d}x\equiv-\int_{-\infty}^{\infty}H(x)\frac{\mathrm{d}f}{\mathrm{d}x}\,\mathrm{d}x. \tag{1.70}$$

对右侧分部积分：

$$\begin{aligned}-\int_{-\infty}^{\infty}H(x)\frac{\mathrm{d}f}{\mathrm{d}x}\,\mathrm{d}x&=-\int_{0}^{\infty}\frac{\mathrm{d}f}{\mathrm{d}x}\,\mathrm{d}x\\&=f(0).\end{aligned} \tag{1.71}$$

于是得到

$$\frac{\mathrm{d}H(x)}{\mathrm{d}x}=\delta(x), \tag{1.72}$$

1.6.1 多维狄拉克函数

三维 δ 函数定义为

$$\delta^{(3)}(\boldsymbol{r})=\delta(x)\delta(y)\delta(z), \tag{1.73}$$

并满足

$$\iiint\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}_0)f(\boldsymbol{r})\,\mathrm{d}V=f(\boldsymbol{r}_0). \tag{1.74}$$

在正交曲线坐标 (u_1,u_2,u_3) 中,

$$\mathrm{d}V=h_1h_2h_3\,\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3,$$

因此定义

$$\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}_0)=\frac{\delta(u_1-u_{1,0})\delta(u_2-u_{2,0})\delta(u_3-u_{3,0})}{h_1h_2h_3}. \tag{1.75}$$

1.7 矢量场理论

1.7.1 亥姆霍兹定理

设矢量场 $\boldsymbol{F}(\boldsymbol{r})$ 定义在整个空间中, 其散度与旋度分别给定为

$$\boldsymbol{\nabla}\cdot\boldsymbol{F}(\boldsymbol{r})=D(\boldsymbol{r}),\qquad\boldsymbol{\nabla}\times\boldsymbol{F}(\boldsymbol{r})=\boldsymbol{C}(\boldsymbol{r}).$$

为了使体积分有意义, 仅要求 $D(\boldsymbol{r})$ 与 $\boldsymbol{C}(\boldsymbol{r})$ 在无穷远处趋于零是不够的, 它们的衰减速度必须足够快。

考虑定义在整个空间上的积分

$$\iiint_{\mathbb{V}}X(\boldsymbol{r})\,\mathrm{d}V,$$

其中 $X(\boldsymbol{r})$ 表示 $D(\boldsymbol{r})$ 或 $\boldsymbol{C}(\boldsymbol{r})$ 的某一分量。

将积分写为球坐标形式,

$$\iiint_{\mathbb{V}} X(\boldsymbol{r})\,\mathrm{d}V = \int_0^\infty \left(\iint_S X(r,\Omega) r^2\,\mathrm{d}\Omega\right)\mathrm{d}r.$$

若存在常数 $A>0$ 及 $R>0$, 使得当 $r>R$ 时对所有方向 Ω 成立

$$|X(r,\Omega)| \geq \frac{A}{r},$$

则有下界估计

$$\iiint_{|\boldsymbol{r}|>R} |X(\boldsymbol{r})|\,\mathrm{d}V \geq A\int_R^\infty r\,\mathrm{d}r \iint_S \mathrm{d}\Omega,$$

该积分显然发散。

类似地, 若

$$|X(r,\Omega)| \geq \frac{A}{r^2},$$

则径向积分包含

$$\int_R^\infty \frac{\mathrm{d}r}{r},$$

从而发散。

因此, 为保证体积分收敛, 必须要求

$$X(\boldsymbol{r}) = o\left(\frac{1}{r^2}\right), \qquad r\to\infty, \tag{1.76}$$

该条件需在角向上一致成立。

因此, 为保证积分收敛, 必须要求

$$D(\boldsymbol{r}),\ \boldsymbol{C}(\boldsymbol{r}) = o\left(\frac{1}{r^2}\right), \qquad r\to\infty. \tag{1.77}$$

这一条件同时也足以保证在无穷远处的曲面积分为零。

设矢量场 $\boldsymbol{F}(\boldsymbol{r})$ 具有散度 $D(\boldsymbol{r})$ 与旋度 $\boldsymbol{C}(\boldsymbol{r})$。若取另一矢量场

$$\boldsymbol{F}'(\boldsymbol{r}) = \boldsymbol{F}(\boldsymbol{r}) + \boldsymbol{G}(\boldsymbol{r}),$$

且 $\boldsymbol{G}(\boldsymbol{r})$ 满足

$$\boldsymbol{\nabla}\cdot\boldsymbol{G} = 0, \qquad \boldsymbol{\nabla}\times\boldsymbol{G} = \boldsymbol{0},$$

则 $\boldsymbol{F}'$ 与 $\boldsymbol{F}$ 具有相同的散度与旋度。

因此, 仅给定散度与旋度并不能唯一确定矢量场。然而可以证明：不存在一个非零的矢量场 (见第三章) $\boldsymbol{G}(\boldsymbol{r})$, 它在整个空间中同时满足

$$\boldsymbol{\nabla}\cdot\boldsymbol{G} = 0, \qquad \boldsymbol{\nabla}\times\boldsymbol{G} = \boldsymbol{0},$$

并且在无穷远处趋于零。

因此, 若进一步要求

$$\boldsymbol{F}(\boldsymbol{r}) \to \boldsymbol{0} \qquad 当 r\to\infty,$$

则满足给定散度与旋度的矢量场解是唯一的。

现在可以严谨地表述亥姆霍兹定理如下：

> 若矢量场 $\boldsymbol{F}(\boldsymbol{r})$ 的散度 $D(\boldsymbol{r})$ 与旋度 $\boldsymbol{C}(\boldsymbol{r})$ 已知, 且二者在 $r\to\infty$ 时均比 $1/r^2$ 衰减得更快, 同时 $\boldsymbol{F}(\boldsymbol{r})$ 在无穷远处趋于零, 则 $\boldsymbol{F}(\boldsymbol{r})$ 被其散度与旋度唯一确定。

1.7.2 势函数

1. 标量势与无旋场 设向量场 $\boldsymbol{F}(\boldsymbol{r})$ 定义在某一区域 $V \subset \mathbb{R}^3$。若存在标量函数 $\phi(\boldsymbol{r})$, 使得

$$\boldsymbol{F} = -\boldsymbol{\nabla}\phi, \tag{1.78}$$

则称 ϕ 为 $\boldsymbol{F}$ 的**标量势函数**, $\boldsymbol{F}$ 称为**保守场**。由旋度的定义可得

$$\boldsymbol{\nabla} \times \boldsymbol{F} = -\boldsymbol{\nabla} \times \boldsymbol{\nabla}\phi = \boldsymbol{0}. \tag{1.79}$$

因此, 任何具有标量势的向量场必为无旋场。

2. 无旋场的势函数存在性 若

$$\boldsymbol{\nabla} \times \boldsymbol{F} = \boldsymbol{0}, \tag{1.80}$$

并且区域 V 是单连通的 (任意闭合曲线可连续收缩为一点), 则由斯托克斯定理可得

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} = \iint_S (\boldsymbol{\nabla} \times \boldsymbol{F}) \cdot \mathrm{d}\boldsymbol{S} = 0, \tag{1.81}$$

线积分与路径无关, 可定义

$$\phi(\boldsymbol{r}) = -\int_{\boldsymbol{r}_0}^{\boldsymbol{r}} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l}, \tag{1.82}$$

从而得到 $\boldsymbol{F} = -\boldsymbol{\nabla}\phi$。

3. 向量势与无散场 若存在向量函数 $\boldsymbol{A}(\boldsymbol{r})$, 使

$$\boldsymbol{F} = \boldsymbol{\nabla} \times \boldsymbol{A}, \tag{1.83}$$

则称 $\boldsymbol{A}$ 为 $\boldsymbol{F}$ 的**向量势**。由恒等式

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) = 0 \tag{1.84}$$

可知, 任何具有向量势的向量场必为无散场。

4. 势函数的不唯一性

- 若 $\boldsymbol{F} = -\boldsymbol{\nabla}\phi$, 则对任意常数 C, $\phi' = \phi + C$ 给出同一向量场。
- 若 $\boldsymbol{F} = \boldsymbol{\nabla} \times \boldsymbol{A}$, 则对任意标量函数 $\chi(\boldsymbol{r})$, $\boldsymbol{A}' = \boldsymbol{A} + \boldsymbol{\nabla}\chi$ 产生相同的 $\boldsymbol{F}$(**规范变换**)。

5. 亥姆霍兹分解 在满足适当衰减条件的情况下, 任意向量场 $\boldsymbol{F}$ 可分解为

$$\boldsymbol{F} = -\boldsymbol{\nabla}\phi + \boldsymbol{\nabla} \times \boldsymbol{A}, \tag{1.85}$$

其中

$$\boldsymbol{\nabla}^2\phi = -\boldsymbol{\nabla} \cdot \boldsymbol{F}, \tag{1.86}$$

$$\boldsymbol{\nabla}^2\boldsymbol{A} = -\boldsymbol{\nabla} \times \boldsymbol{F}. \tag{1.87}$$

这就是亥姆霍兹定理在势函数语言下的表达。

1.8 习题

1.8.1 证明: $\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c})+\boldsymbol{c}\times(\boldsymbol{a}\times\boldsymbol{b})+\boldsymbol{b}\times(\boldsymbol{c}\times\boldsymbol{a})=0$

$$\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c})+\boldsymbol{c}\times(\boldsymbol{a}\times\boldsymbol{b})+\boldsymbol{b}\times(\boldsymbol{c}\times\boldsymbol{a})=-(\boldsymbol{b}\cdot\boldsymbol{a})\boldsymbol{c}+(\boldsymbol{a}\cdot\boldsymbol{b})\boldsymbol{c}-(\boldsymbol{a}\cdot\boldsymbol{c})\boldsymbol{b}+(\boldsymbol{b}\cdot\boldsymbol{c})\boldsymbol{a}-(\boldsymbol{c}\cdot\boldsymbol{b})\boldsymbol{a}+(\boldsymbol{a}\cdot\boldsymbol{b})\boldsymbol{c}=0$$

1.8.2 证明: $\boldsymbol{Ra}\cdot\boldsymbol{Rb}=\boldsymbol{a}\cdot\boldsymbol{b}$, 其中 $\boldsymbol{R}$ 为旋转矩阵

$$\boldsymbol{Ra}\cdot\boldsymbol{Rb}=(\boldsymbol{Ra})^T(\boldsymbol{Rb})=\boldsymbol{a}^T\boldsymbol{R}^T\boldsymbol{Rb}=\boldsymbol{a}^T\boldsymbol{R}^{-1}\boldsymbol{Rb}=\boldsymbol{a}^T\boldsymbol{b}=\boldsymbol{a}\cdot\boldsymbol{b}$$

1.8.3 在坐标逆变换 $(x'=-x,\ y'=-y,\ z'=-z)$ 下, 两个矢量的叉乘是如何变换的?

$$\boldsymbol{a}'\times\boldsymbol{b}'=\begin{pmatrix}\boldsymbol{e}_i & \boldsymbol{e}_j & \boldsymbol{e}_k\\ a_i' & a_j' & a_k'\\ b_i' & b_j' & b_k'\end{pmatrix}=\begin{pmatrix}\boldsymbol{e}_i & \boldsymbol{e}_j & \boldsymbol{e}_k\\ -a_i & -a_j & -a_k\\ -b_i & -b_j & -b_k\end{pmatrix}=\boldsymbol{a}\times\boldsymbol{b}$$

1.8.4 在坐标逆变换下, 标量三重积 $(\boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b}))$ 是如何变换的?

$$\boldsymbol{c}'\cdot(\boldsymbol{a}'\times\boldsymbol{b}')=\boldsymbol{c}'\cdot(\boldsymbol{a}\times\boldsymbol{b})=-\boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b})$$

1.8.5 $\boldsymbol{n}$ 是固定点 (x',y',z') 到点 (x,y,z) 的间隔矢量, n 是它的长度, 证明:

(a) $\boldsymbol{\nabla}n^2=2\boldsymbol{n}$.

(b) $\boldsymbol{\nabla}\frac{1}{n}=-\frac{\boldsymbol{n}}{n^2}$.

(c) $\boldsymbol{\nabla}n^a=a\boldsymbol{n}n^{a-2}(a\neq 0)$.

(a) $\boldsymbol{\nabla}n^2=\frac{\partial n^2}{\partial e_i}\boldsymbol{e}_i=2n\frac{\partial n}{\partial e_i}\boldsymbol{e}_i=2\boldsymbol{n}$

(c) $\boldsymbol{\nabla}n^a=\frac{\partial n^a}{\partial e_i}\boldsymbol{e}_i=an^{-1}\frac{\partial n}{\partial e_i}\boldsymbol{e}_i=a\boldsymbol{n}n^{a-2}$

1.8.6 函数 f 只依赖平面坐标 y,z。令旋转后的坐标记为 $\begin{pmatrix}y'\\ z'\end{pmatrix}=\begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}y\\ z\end{pmatrix}$, 求证 $\boldsymbol{\nabla}f'=\begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\frac{\partial f}{\partial y}\\ \frac{\partial f}{\partial z}\end{pmatrix}$

$$\begin{pmatrix}y\\ z\end{pmatrix}=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}y'\\ z'\end{pmatrix}$$

$$\frac{\partial f}{\partial y'}=\frac{\partial f}{\partial y}\frac{\partial y}{\partial y'}+\frac{\partial f}{\partial z}\frac{\partial z}{\partial y'}=\frac{\partial f}{\partial y}\cos\theta+\frac{\partial f}{\partial z}\sin\theta$$

$$\frac{\partial f}{\partial z'}=\frac{\partial f}{\partial y}\frac{\partial y}{\partial z'}+\frac{\partial f}{\partial z}\frac{\partial z}{\partial z'}=-\frac{\partial f}{\partial y}\sin\theta+\frac{\partial f}{\partial z}\cos\theta$$

$$\begin{pmatrix}\dfrac{\partial f}{\partial y'}\\ \dfrac{\partial f}{\partial z'}\end{pmatrix}=\begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\dfrac{\partial f}{\partial y}\\ \dfrac{\partial f}{\partial z}\end{pmatrix}$$

1.8.7 计算 $(\boldsymbol{\nabla T})\times(\boldsymbol{\nabla S})$

$$
\begin{aligned}
(\boldsymbol{\nabla T})\times(\boldsymbol{\nabla S}) &= \left(\varepsilon_{ijk}\frac{\partial T_j}{\partial x_i}\boldsymbol{e}_k\right)\times\left(\varepsilon_{lmn}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_n\right)\\
&= \varepsilon_{opq}\varepsilon_{ijo}\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q\\
&= \varepsilon_{pqo}\varepsilon_{ijo}\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q\\
&= (\delta_{ip}\delta_{jq}-\delta_{iq}\delta_{jp})\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q\\
&= \delta_{ip}\delta_{jq}\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q-\delta_{iq}\delta_{jp}\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q\\
&= \frac{\partial T_j}{\partial x_i}\varepsilon_{lmi}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_j-\frac{\partial T_j}{\partial x_i}\varepsilon_{lmj}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_i\\
&= \frac{\partial T_i}{\partial x_j}\varepsilon_{lmj}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_i-\frac{\partial T_j}{\partial x_i}\varepsilon_{lmj}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_i\\
&= \left(\frac{\partial T_i}{\partial x_j}-\frac{\partial T_j}{\partial x_i}\right)\varepsilon_{lmj}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_i
\end{aligned}
$$

1.8.8 证明 $\iint_S f(\boldsymbol{\nabla}\times\boldsymbol{a})\cdot\mathrm{d}\boldsymbol{S}=\iint_S[\boldsymbol{a}\times(\boldsymbol{\nabla}f)]\cdot\mathrm{d}\boldsymbol{S}+\oint_{\partial S}f\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l}$

$$
\begin{aligned}
\oint_{\partial S}f\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l} &= \iint_S[\boldsymbol{\nabla}\times(\boldsymbol{f}\boldsymbol{a})]\cdot\mathrm{d}\boldsymbol{S}\\
&= \iint_S[f(\boldsymbol{\nabla}\times\boldsymbol{a})+(\boldsymbol{\nabla}f)\times\boldsymbol{a}]\cdot\mathrm{d}\boldsymbol{S}\\
\iint_S f(\boldsymbol{\nabla}\times\boldsymbol{a})\cdot\mathrm{d}\boldsymbol{S} &= \iint_S[\boldsymbol{a}\times(\boldsymbol{\nabla}f)]\cdot\mathrm{d}\boldsymbol{S}+\oint_{\partial S}f\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l}
\end{aligned}
$$

1.8.9 证明 $\iiint_V\boldsymbol{b}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}\tau=\iiint_V\boldsymbol{a}\cdot(\boldsymbol{\nabla}\times\boldsymbol{b})\,\mathrm{d}\tau+\iint(\boldsymbol{a}\times\boldsymbol{b})\cdot\mathrm{d}\boldsymbol{S}_{\partial V}$

$$
\begin{aligned}
\iint_{\partial V}(\boldsymbol{a}\times\boldsymbol{b})\cdot\mathrm{d}\boldsymbol{S} &= \iiint_V\boldsymbol{\nabla}\cdot(\boldsymbol{a}\times\boldsymbol{b})\,\mathrm{d}V\\
&= \iiint_V[\boldsymbol{b}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})-\boldsymbol{a}\cdot(\boldsymbol{\nabla}\times\boldsymbol{b})]\,\mathrm{d}V\\
\iiint_V\boldsymbol{b}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}\tau &= \iiint_V\boldsymbol{a}\cdot(\boldsymbol{\nabla}\times\boldsymbol{b})\,\mathrm{d}\tau+\iint_{\partial V}(\boldsymbol{a}\times\boldsymbol{b})\cdot\mathrm{d}\boldsymbol{S}
\end{aligned}
$$

1.8.10 设 $f=f(r)$, 求 $\boldsymbol{\nabla}^2[f(r)]$

$$
\begin{aligned}
\boldsymbol{\nabla}^2[f(r)] &= \frac{1}{sr}\frac{\partial}{\partial r}\left(sr\frac{\partial f}{\partial r}\right)+\frac{1}{s^2}\frac{\partial}{\partial \phi}\left(\frac{\partial f}{\partial \phi}\right)+\frac{1}{sr^2}\frac{\partial}{\partial \theta}\left(s\frac{\partial f}{\partial \theta}\right)\\
&= \frac{1}{sr}\frac{\partial}{\partial r}\left(sr\frac{\partial f}{\partial r}\right)\\
&= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right)\\
&= \frac{2}{r}\frac{\partial f}{\partial r}+\frac{\partial^2 f}{\partial r^2}
\end{aligned}
\tag{1.88}
$$

1.8.11 证明:$\iiint_V \boldsymbol{\nabla}\times\boldsymbol{a}\,\mathrm{d}V=\oiint \mathrm{d}\boldsymbol{S}\times\boldsymbol{a}$

取任意常向量 $\boldsymbol{c}$。利用式1.39得 $\boldsymbol{\nabla}\times\boldsymbol{c}=\boldsymbol{0}$, 因此

$$\boldsymbol{\nabla}\cdot(\boldsymbol{a}\times\boldsymbol{c})=\boldsymbol{c}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a}).$$

应用高斯定理,

$$\iiint_V \boldsymbol{\nabla}\cdot(\boldsymbol{a}\times\boldsymbol{c})\,\mathrm{d}V=\oiint_{\partial V}(\boldsymbol{a}\times\boldsymbol{c})\cdot\mathrm{d}\boldsymbol{S}=\iiint_V \boldsymbol{c}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}V.$$

利用式1.12得

$$\boldsymbol{c}\cdot\iiint_V(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}V=\boldsymbol{c}\cdot\oiint_{\partial V}\mathrm{d}\boldsymbol{S}\times\boldsymbol{a}.$$

由于上述等式对任意常向量 $\boldsymbol{c}$ 成立, 必有

$$\iiint_V(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}V=\oiint_{\partial V}\mathrm{d}\boldsymbol{S}\times\boldsymbol{a}.$$

1.8.12 证明:$\iint_S \mathrm{d}\boldsymbol{S}\times\boldsymbol{\nabla}f=\oint_{\partial S} f\,\mathrm{d}\boldsymbol{l}$

取任意常向量 $\boldsymbol{c}$。利用式1.28得

$$\iint_S(\boldsymbol{\nabla}f\times\boldsymbol{c})\cdot\mathrm{d}\boldsymbol{S}=\iint_S(\boldsymbol{\nabla}\times(f\boldsymbol{c}))\cdot\mathrm{d}\boldsymbol{S}.$$

由斯托克斯定理

$$\iint_S(\boldsymbol{\nabla}\times\boldsymbol{A})\cdot\mathrm{d}\boldsymbol{S}=\oint_{\partial S}\boldsymbol{A}\cdot\mathrm{d}\boldsymbol{l},$$

取 $\boldsymbol{A}=f\boldsymbol{c}$, 得

$$\iint_S(\boldsymbol{\nabla}\times(f\boldsymbol{c}))\cdot\mathrm{d}\boldsymbol{S}=\oint_{\partial S}(f\boldsymbol{c})\cdot\mathrm{d}\boldsymbol{l}=\oint_{\partial S}f\,\boldsymbol{c}\cdot\mathrm{d}\boldsymbol{l}.$$

又由混合积恒等式

$$\boldsymbol{c}\cdot(\mathrm{d}\boldsymbol{S}\times\boldsymbol{\nabla}f)=(\boldsymbol{\nabla}f\times\boldsymbol{c})\cdot\mathrm{d}\boldsymbol{S},$$

所以

$$\boldsymbol{c}\cdot\iint_S\mathrm{d}\boldsymbol{S}\times\boldsymbol{\nabla}f=\boldsymbol{c}\cdot\oint_{\partial S}f\ \mathrm{d}\boldsymbol{l}.$$

由于上述等式对任意常向量 $\boldsymbol{c}$ 都成立, 必有

$$\iint_S\mathrm{d}\boldsymbol{S}\times\boldsymbol{\nabla}f=\oint_{\partial S}f\ \mathrm{d}\boldsymbol{l}.$$

1.8.13 证明:$\oiint_{\partial S}(\boldsymbol{a}\times\boldsymbol{r})\cdot\mathrm{d}\boldsymbol{l}=2\iint_S\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S}$

$$
\begin{aligned}
\oiint_{\partial S}(\boldsymbol{a}\times\boldsymbol{r})\cdot\mathrm{d}\boldsymbol{l}&=\iint_S[\boldsymbol{\nabla}\times(\boldsymbol{a}\times\boldsymbol{r})]\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S[\boldsymbol{\nabla}\times(\varepsilon_{ijk}a_ix_j\boldsymbol{e}_k)]\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(\varepsilon_{lmn}\varepsilon_{ijm}\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n)\cdot\mathrm{d}\boldsymbol{S}\\
&=-\iint_S(\varepsilon_{lnm}\varepsilon_{ijm}\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n)\cdot\mathrm{d}\boldsymbol{S}\\
&=-\iint_S(\delta_{il}\delta_{jn}-\delta_{in}\delta_{jl})(\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n)\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(\delta_{in}\delta_{jl}\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n-\delta_{il}\delta_{jn}\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n)\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(\frac{\partial a_ix_j}{\partial x_j}\boldsymbol{e}_i-\frac{\partial a_ix_j}{\partial x_i}\boldsymbol{e}_j)\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(\frac{\partial a_jx_i}{\partial x_i}\boldsymbol{e}_j-\frac{\partial a_ix_j}{\partial x_i}\boldsymbol{e}_j)\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(a_j\frac{\partial x_i}{\partial x_i}+x_i\frac{\partial a_j}{\partial x_i}-a_i\frac{\partial x_j}{\partial x_i}-x_j\frac{\partial a_i}{\partial x_i})\boldsymbol{e}_j\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(3a_j+x_i\frac{\partial a_j}{\partial x_i}-a_i\delta_{ij}-x_j\frac{\partial a_i}{\partial x_i})\boldsymbol{e}_j\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S 2a_j\boldsymbol{e}_j\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S}
\end{aligned}
$$

1.8.14 将下列电荷分布表示成电荷密度函数 ρ:1. 电荷量 Q 均匀分布在半径为 R 的球面上; 2. 电荷均匀分布在半径为 R 的柱面上, 单位长度的电荷量为 λ; 3. 电荷量均匀分布在半径为 R 的平面圆盘上

$$
\begin{aligned}
&1.\rho=\delta(R\boldsymbol{e}_r-\boldsymbol{r})\frac{Q}{4\pi R^2}\\
&2.\rho=\delta(R\boldsymbol{e}_s-\boldsymbol{s})\frac{\lambda}{2\pi R}\\
&3.\rho=\delta(\boldsymbol{x}_3)H(x+R)H(-x-R)\frac{\lambda}{\pi R^2}
\end{aligned}
$$

1.8.15 对函数 $\boldsymbol{a} = r^2\cos\theta\boldsymbol{r} + r^2\cos\phi\boldsymbol{\theta} - r^2\cos\theta\sin\phi\boldsymbol{\phi}$ 验证散度定理, 体积为半径为 R 在第一卦限的 $1/8$ 球体

$$\boldsymbol{\nabla}\cdot\boldsymbol{a} = 2r\cos\theta - r^2\cos\theta\cos\phi$$

$$\begin{aligned}
\int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R\boldsymbol{\nabla}\cdot\boldsymbol{a}\,\mathrm{d}r &= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R\left(2r\cos\theta - r^2\cos\theta\cos\phi\right)\mathrm{d}r \\
&= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\left(r^2\cos\theta - \frac{r^3}{3}\cos\theta\cos\phi\right)\Big|_0^R \\
&= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}\left(R^2\cos\theta - \frac{R^3}{3}\cos\theta\cos\phi\right)\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\left(R^2\sin\theta - \frac{R^3}{3}\sin\theta\cos\phi\right)\Big|_0^{\frac{\pi}{2}} \\
&= \int_0^{\frac{\pi}{2}}\left(R^2 - \frac{R^3}{3}\cos\phi\right)\mathrm{d}\phi \\
&= \left(R^2\phi - \frac{R^3}{3}\sin\phi\right)\Big|_0^{\frac{\pi}{2}} \\
&= \frac{\pi}{2}R^2 - \frac{R^3}{3}
\end{aligned}$$

计算 XY 平面:

$$\begin{aligned}
\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R a_\phi\,\mathrm{d}r &= -\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R r^2\cos\theta\sin\phi\,\mathrm{d}r \\
&= -\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R r^2\cos\theta\,\mathrm{d}r \\
&= -\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\,\frac{r^3}{3}\cos\theta\Big|_0^R \\
&= -\int_0^{\frac{\pi}{2}}\frac{R^3}{3}\cos\theta\,\mathrm{d}\theta \\
&= -\frac{R^3}{3}
\end{aligned}$$

计算 XZ 平面:

$$\begin{aligned}
\int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^R a_\theta\,\mathrm{d}r &= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^R r^2\cos\phi\,\mathrm{d}r \\
&= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\,\frac{r^3}{3}\cos\phi\Big|_0^R \\
&= \int_0^{\frac{\pi}{2}}\frac{R^3}{3}\cos\phi\,\mathrm{d}\phi \\
&= \frac{R^3}{3}
\end{aligned}$$

计算 YZ 平面:

$$\int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^R a_\theta\,\mathrm{d}r = \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^R r^2\cos\phi\,\mathrm{d}r = \frac{R^3}{3}$$

1.8.16 对函数 $\boldsymbol{a}=bx_2\boldsymbol{e}_1+cx_1\boldsymbol{e}_2$ 验证斯托克斯定理, 面的边界线选为处在 XY 平面, 半径为 R, 圆心在原点的圆周线。

$$\boldsymbol{\nabla}\times\boldsymbol{a}=\boldsymbol{\nabla}\times(by\boldsymbol{e}_1+cx\boldsymbol{e}_2)=\begin{pmatrix}\boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3\\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3}\\ bx_2 & cx_1 & 0\end{pmatrix}=(c-b)\boldsymbol{e}_3$$

$$\iint\boldsymbol{\nabla}\times\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l}=\iint(c-b)\boldsymbol{e}_3\cdot\mathrm{d}\boldsymbol{S}=2\pi R(c-b)$$

$$\begin{aligned}\oint(bR\sin\phi\boldsymbol{e}_1+cR\cos\phi\boldsymbol{e}_2)\cdot\mathrm{d}\boldsymbol{l}=&\int_0^{2\pi}(bR\sin\phi\boldsymbol{e}_1+cR\cos\phi\boldsymbol{e}_2)\cdot(-R\sin\phi\boldsymbol{e}_1\ +R\cos\phi\boldsymbol{e}_2\ )\,\mathrm{d}\phi\\ =&\int_0^{2\pi}(-bR^2\sin^2\phi+cR^2\cos^2\phi)\,\mathrm{d}\phi\\ =&2\pi R(c-b)\end{aligned}$$

1.8.17 对函数 $\boldsymbol{a}=r\cos^2\theta\boldsymbol{r}-r\sin\theta\cos\theta\boldsymbol{\theta}+3r\boldsymbol{\phi}$ 验证斯托克斯定理。

$$\begin{aligned}&\boldsymbol{\nabla}\times\boldsymbol{a}\\ =&\frac{1}{sr}\left(r\frac{\partial a_\theta}{\partial\phi}-s\frac{\partial a_\phi}{\partial\theta}\right)\boldsymbol{r}+\frac{1}{r}\left(\frac{\partial a_r}{\partial\theta}-\frac{\partial ra_\theta}{\partial r}\right)\boldsymbol{\phi}+\frac{1}{s}\left(\frac{\partial sa_\phi}{\partial r}-\frac{\partial a_r}{\partial\phi}\right)\boldsymbol{\theta}\\ =&\frac{1}{sr}\left(r\frac{\partial r\sin\theta\cos\theta}{\partial\phi}-s\frac{\partial 3r}{\partial\theta}\right)\boldsymbol{r}+\frac{1}{r}\left(\frac{\partial r\cos^2\theta}{\partial\theta}-\frac{\partial rr\sin\theta\cos\theta}{\partial r}\right)\boldsymbol{\phi}+\frac{1}{s}\left(\frac{\partial s3r}{\partial r}-\frac{\partial r\cos^2\theta}{\partial\phi}\right)\boldsymbol{\theta}\\ =&\frac{1}{s}\left(6r\sin\theta\right)\boldsymbol{\theta}\\ =&6\boldsymbol{\theta}\end{aligned}$$

计算 XY 平面:

$$\begin{aligned}\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R a_\phi\,\mathrm{d}r=&\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R 6\,\mathrm{d}r\\ =&\frac{3}{2}R\end{aligned}$$

1.8.18 对函数 $\boldsymbol{a} = r^2\sin^2\theta\boldsymbol{r} + 4r^2\cos\theta\boldsymbol{\theta} + r^2\tan\theta\boldsymbol{\phi}$ 验证散度定理。

$$
\begin{aligned}
\boldsymbol{\nabla}\cdot\boldsymbol{a} &= \frac{1}{sr}(\frac{\partial sra_r}{\partial r} + r\frac{\partial a_\phi}{\partial\phi} + s\frac{\partial\theta}{\partial\theta}) \\
&= \frac{1}{sr}(\frac{\partial srr^2\sin^2\theta}{\partial r} + r\frac{\partial 4r^2\cos\theta}{\partial\phi} + s\frac{\partial r^2\tan\theta}{\partial\theta}) \\
&= \frac{1}{sr}(4sr^2\sin^2\theta + sr^2\frac{1}{\cos^2\theta}) \\
&= 4r\sin^2\theta + r\frac{1}{\cos^2\theta}
\end{aligned}
$$

$$
\begin{aligned}
&\int_0^R r^2\,\mathrm{d}r\int_0^{2\pi}\mathrm{d}\phi\int_0^{\frac{\pi}{6}}(4r\sin^2\theta + r\frac{1}{\cos^2\theta})\sin\theta\,\mathrm{d}\theta \\
=&\int_0^R 2\pi r^3\,\mathrm{d}r\int_0^{\frac{\pi}{6}}(4\sin^2\theta + \frac{1}{\cos^2\theta})\,\mathrm{d}\theta \\
=&\int_0^R 2\pi r^3\ (2\theta - \sin 2\theta + \tan\theta)|_0^{\frac{\pi}{6}}\ \mathrm{d}r \\
=&\int_0^R 2\pi r^3(\frac{\pi}{3} - \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2})\,\mathrm{d}r \\
=&\int_0^R 2\pi r^3\frac{\pi}{3}\,\mathrm{d}r \\
=&2\pi R^4\frac{\pi}{12}
\end{aligned}
$$

1.8.19 证明 $\iiint_V\left[f\boldsymbol{\nabla}^2 g + (\boldsymbol{\nabla}f)\cdot(\boldsymbol{\nabla}g)\right]\mathrm{d}V = \oiint_S(f\boldsymbol{\nabla}g)\cdot\mathrm{d}S$

$$
\begin{aligned}
\iiint_V(\boldsymbol{\nabla}\cdot\boldsymbol{a})\,\mathrm{d}V &= \iint_{\partial V}\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S} \\
\iiint_V[\boldsymbol{\nabla}\cdot(f\boldsymbol{\nabla}g)]\,\mathrm{d}V &= \iint_{\partial V}(f\boldsymbol{\nabla}g)\cdot\mathrm{d}\boldsymbol{S} \\
\iiint_V\left[f\boldsymbol{\nabla}^2 g + (\boldsymbol{\nabla}f)\cdot(\boldsymbol{\nabla}g)\right]\mathrm{d}V &= \iint_{\partial V}(f\boldsymbol{\nabla}g)\cdot\mathrm{d}\boldsymbol{S}
\end{aligned}
$$

1.8.20 求 $\boldsymbol{a}=r^n\boldsymbol{e}_r$ 的散度

$$
\begin{aligned}
\boldsymbol{\nabla}\cdot\boldsymbol{a} &= \frac{\partial a_i}{\partial x_i}\\
&= \frac{\partial x_i r^{n-1}}{\partial x_i}\\
&= 3r^{n-1}+x_i(n-1)r^{n-2}\frac{\partial r}{\partial x_i}\\
&= 3r^{n-1}+x_i(n-1)r^{n-2}\frac{x_i}{r}\\
&= 3r^{n-1}+(n-1)r^{n-1}\\
&= (n+2)r^{n-1}
\end{aligned}
$$

$$
\iint_{\partial V}\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S}=\iint_{\partial V}r^n\boldsymbol{e}_r\cdot\mathrm{d}\boldsymbol{S}=4\pi r^2r^n=
$$

1.8.21 求 $\boldsymbol{a}=r^n\boldsymbol{e}_r$ 的旋度

$$
\begin{aligned}
\boldsymbol{\nabla}\times\boldsymbol{a} &= \varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_k\\
&= \varepsilon_{ijk}\frac{\partial r^{n-1}x_j}{\partial x_i}\boldsymbol{e}_k\\
&= \varepsilon_{ijk}x_j\frac{\partial r^{n-1}}{\partial x_i}\boldsymbol{e}_k\\
&= (n-1)\varepsilon_{ijk}x_jr^{n-2}\frac{\partial r}{\partial x_i}\boldsymbol{e}_k\\
&= (n-1)\varepsilon_{ijk}x_jr^{n-2}\frac{x_i}{r}\boldsymbol{e}_k\\
&= (n-1)r^{n-3}\varepsilon_{ijk}x_ix_j\boldsymbol{e}_k\\
&= 0
\end{aligned}
$$

1.8.22 以匀角速绕轴转动的抛物线形金属丝，其方程为 $x^2=4ay$。一质量为 m 的小环套在此金属丝上，可沿着金属丝无摩擦滑动。求小环在 x 方向的运动微分方程。

$$
\begin{aligned}
\frac{\partial\dfrac{mx^2\omega^2}{2}+mgy+\dfrac{m\dot{y}^2}{2}+\dfrac{m\dot{x}^2}{2}}{\partial t}&=0\\
\frac{\partial\dfrac{mx^2\omega^2}{2}+mg\dfrac{x^2}{4a}+\dfrac{mx^2\dot{x}^2}{8a^2}+\dfrac{m\dot{x}^2}{2}}{\partial t}&=0\\
x\dot{x}\omega^2+gx\frac{\dot{x}}{2a}+\frac{x\dot{x}^3}{4a^2}+\frac{x^2\dot{x}\ddot{x}}{4a^2}+mx\dot{x}&=0\\
x\omega^2+\frac{gx}{2a}+\frac{x\dot{x}^2}{4a^2}+\frac{x^2\ddot{x}}{4a^2}+mx&=0
\end{aligned}
$$

2 静电学

2.1 电场

2.1.1 库仑定律

设一个静止点电荷 q 距检验电荷 Q 的距离为 r, 那么它作用在 Q 上的力是

$$\boldsymbol{F} = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{r^3}\boldsymbol{r} \tag{2.1}$$

常数 ε_0 称为真空介电常数, $\varepsilon_0 = 8.85 \times 10^{-12}\ \dfrac{\mathrm{C}^2}{\mathrm{N}\cdot\mathrm{m}^2}$

2.1.2 电场

$$\boldsymbol{E} \equiv \frac{\boldsymbol{F}}{Q} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r} \tag{2.2}$$

2.1.3 电场强度通量

$\boldsymbol{\Phi}_E$ 为电场强度通量

$$\boldsymbol{\Phi}_E \equiv \iint_S \boldsymbol{E}\cdot\mathrm{d}\boldsymbol{S} \tag{2.3}$$

2.1.4 高斯定理

ρ 为电荷密度

$$\boldsymbol{\nabla}\cdot\boldsymbol{E} = \frac{1}{\varepsilon_0}\rho \tag{2.4}$$

2.1.5 两个带点为 q 的电荷, 相距 d 放置, 求垂直于连线中点且距离为 x_2 处的电场, 如果把一个换成 $-q$ 会怎样? 当 $y >> d$ 时会怎样?

$$\begin{aligned}\frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r}_1 + \frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r}_2 &= 2\frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}x_2\boldsymbol{e}_2 \\ &= \frac{1}{2\pi\varepsilon_0}\frac{qx_2}{r^3}\boldsymbol{e}_2 \\ &= \frac{1}{2\pi\varepsilon_0}\frac{qx_2}{(x_2^2+\frac{d^2}{4})^{\frac{3}{2}}}\boldsymbol{e}_2\end{aligned}$$

$$\begin{aligned}\frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r}_1 + \frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r}_2 &= 2\frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\frac{d}{2}\boldsymbol{e}_1 \\ &= \frac{1}{4\pi\varepsilon_0}\frac{qd}{r^3}\boldsymbol{e}_1 \\ &\approx \frac{1}{4\pi\varepsilon_0}\frac{qd}{x_2^3}\boldsymbol{e}_1\end{aligned}$$

2.1.6 一个长度为 $2L$ 的细杆均匀带电, 电荷线密度为 λ, 求垂直于杆且与杆中心距离为 x_2 处的电场

$$
\begin{aligned}
\int_{-L}^{L} \frac{1}{4\pi\varepsilon_0}\frac{\lambda}{r^3}\boldsymbol{r}\,\mathrm{d}x &= \int_{-L}^{L} \frac{1}{4\pi\varepsilon_0}\frac{\lambda}{r^3}x_2\boldsymbol{e}_2\,\mathrm{d}x \\
&= \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\int_{-L}^{L}\frac{x_2}{r^3}\,\mathrm{d}x \\
&= \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\int_{-L}^{L}\frac{x_2}{(x^2+x_2^2)^{\frac{3}{2}}}\,\mathrm{d}x \\
&\xlongequal{x=x_2\tan\theta} \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\int_{-\arctan\frac{L}{x_2}}^{\arctan\frac{L}{x_2}}\frac{x_2}{(x_2^2\tan\theta^2+x_2^2)^{\frac{3}{2}}}\,\mathrm{d}x_2\tan\theta \\
&= \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\int_{-\arctan\frac{L}{x_2}}^{\arctan\frac{L}{x_2}}\frac{x_2^2\cos^3\theta}{x_2^3\cos^2\theta}\,\mathrm{d}\theta \\
&= \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\int_{-\arctan\frac{L}{x_2}}^{\arctan\frac{L}{x_2}}\frac{\cos\theta}{x_2^2}\,\mathrm{d}\theta \\
&= \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\left.\frac{\sin\theta}{x_2}\right|_{-\arctan\frac{L}{x_2}}^{\arctan\frac{L}{x_2}} \\
&= \frac{\lambda}{2\pi\varepsilon_0 x_2}\boldsymbol{e}_2\frac{L}{\sqrt{L^2+x_2^2}}
\end{aligned}
$$

2.1.7 一个长度为 L 的细杆均匀带电, 电荷线密度为 λ, 求杆一端上方距杆 y 处的电场

$$
\begin{aligned}
\int_0^L \frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0 r^2}\,\mathrm{d}x_2 &= \int_0^L \frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0(x_2+y)^2}\,\mathrm{d}x_2 \\
&= -\left.\frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0(x_2+y)}\right|_0^L \\
&= \frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0(L+y)} - \frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0 y} \\
&= \frac{\lambda\boldsymbol{e}_y L}{4\pi\varepsilon_0(L+y)y}
\end{aligned}
$$

2.1.8 一个边长为 $2L$ 的正方形线框均匀带电, 电荷线密度为 λ, 求线框中心上方距 x_3 处的电场

$$
4\frac{\lambda}{2\pi\varepsilon_0\sqrt{L^2+x_3^2}}\boldsymbol{e}_3\frac{L}{2\sqrt{2L^2+x_3^2}}\frac{x_3}{x_3\sqrt{L^2+x_3^2}^2} = \frac{\lambda}{\pi\varepsilon_0\sqrt{2L^2+x_3^2}}\boldsymbol{e}_3\frac{2Lx_3}{L^2+x_3^2}
$$

2.1.9 一个半径为 r 的圆线框均匀带电, 电荷线密度为 λ, 求线框中心上方距 x_3 处的电场

$$2\pi r\frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2+r^2}}=\frac{\lambda}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3 r}{\sqrt{x_3^2+r^2}^3}$$

2.1.10 一个半径为 r 的圆片均匀带电, 电荷面密度为 σ, 求圆片中心上方距 x_3 处的电场

$$\begin{aligned}\int_0^r\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3x_1}{\sqrt{x_3^2+x_1^2}^3}\,\mathrm{d}x_1&=-\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2+x_1^2}}\bigg|_0^r\\&=\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2}}-\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2+r^2}}\\&=\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3-\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2+r^2}}\end{aligned}$$

2.1.11 一个半径为 r 的球面均匀带电, 电荷面密度为 σ, 求圆球上方距 d 处的电场 (分 $d>r$ 和 $d<r$ 讨论)

$$\begin{aligned}\int_\pi^0\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{(x_3+d)\sqrt{x_1^2+x_2^2}}{\sqrt{(x_3+d)^2+x_1^2+x_2^2}^3}\,\mathrm{d}r\phi&=\int_\pi^0\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{(r\cos\phi+d)\sqrt{x_1^2+x_2^2}}{\sqrt{(r\cos\phi+d)^2+x_1^2+x_2^2}^3}\,\mathrm{d}r\phi\\&=\int_\pi^0\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{(r\cos\phi+d)r\sin\phi}{\sqrt{r^2+2rd\cos\phi+d^2}^3}\,\mathrm{d}r\phi\\&=\int_0^\pi\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{(r\cos\phi+d)r^2\sin\phi}{\sqrt{r^2+2rd\cos\phi+d^2}^3}\,\mathrm{d}\phi\end{aligned}$$

2.1.12 在某个区域电场可以写为 $\boldsymbol{E}=kr^3\boldsymbol{e_r}$, 求电荷密度和包含在半径为 R, 球心在原点的闭合球面内的总电荷

$$\rho=\varepsilon_0\boldsymbol{\nabla}\cdot\boldsymbol{E}=\varepsilon_0\frac{1}{sr}\frac{\partial srE_r}{\partial r}=\varepsilon_0\frac{1}{sr}\frac{\partial srr^3}{\partial r}=\varepsilon_0\frac{1}{sr}5sr^4=5\varepsilon_0r^3k$$

$$4\pi\int_0^R5\varepsilon_0r^3\,\mathrm{d}r=5\pi\varepsilon_0R^4k$$

2.2 电势

2.2.1 定义

$$V(\boldsymbol{r})\equiv-\int_O^r\boldsymbol{E}\cdot\mathrm{d}\boldsymbol{l}\tag{2.5}$$

其中 O 为预先设置的标准参考点, 通常为无限远处

2.2.2 一个半径为 R 的均匀带电球体, 总电荷为 q, 求电势 $V(\boldsymbol{r})$

$r \geq R$:

$$\int_r^\infty \frac{q}{4\pi\varepsilon_0 x^2}\,\mathrm{d}x = -\frac{q}{4\pi\varepsilon_0 x}\bigg|_r^\infty = \frac{q}{4\pi\varepsilon_0 r}$$

$r < R$:

$$\begin{aligned}\frac{q}{4\pi\varepsilon_0 R} + \int_r^R \frac{q}{4\pi\varepsilon_0 x^2}\frac{x^3}{R^3}\,\mathrm{d}x &= \frac{q}{4\pi\varepsilon_0 R} + \frac{qx^2}{4\pi\varepsilon_0 R^3}\bigg|_r^R \\ &= \frac{q}{4\pi\varepsilon_0 R} + \frac{qR^2}{4\pi\varepsilon_0 R^3} - \frac{qr^2}{4\pi\varepsilon_0 R^3} \\ &= \frac{q}{2\pi\varepsilon_0 R} - \frac{qr^2}{4\pi\varepsilon_0 R^3}\end{aligned}$$

2.2.3 一条均匀带电的无限长直线, 电荷线密度为 λ, 求电势 $V(\boldsymbol{r})$

$$V = \int_r^\infty \frac{\lambda}{2\pi r}\,\mathrm{d}x = \frac{\lambda}{2\pi}\ln(x)\bigg|_r^\infty = \frac{1}{2\pi}\ln(r)$$

2.2.4 一个长度为 $2L$ 的细杆均匀带电, 电荷线密度为 λ, 求垂直于杆且与杆中心距离为 h 处的电势

$$\begin{aligned}V &= \int_h^\infty \frac{\lambda}{4\pi\varepsilon_0}\frac{2L}{x_2\sqrt{L^2+x_2^2}}\,\mathrm{d}x_2 \\ &= -\frac{\lambda}{2\pi\varepsilon_0}\ln\left(\frac{\sqrt{x^2+L^2}+L}{x}\right)\bigg|_h^\infty \\ &= \frac{\lambda}{2\pi\varepsilon_0}\ln\left(\frac{\sqrt{h^2+L^2}+L}{h}\right)\end{aligned}$$

2.2.5 一个半径为 r 的圆片均匀带电, 电荷面密度为 σ, 求圆片中心上方距 h 处的电势

$$\begin{aligned}V &= \int_h^\infty \left(\frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0}\frac{x_3}{\sqrt{x_3^2+r^2}}\right)\mathrm{d}x_3 \\ &= \frac{\sigma}{2\varepsilon_0}x_3 - \frac{\sigma}{2\varepsilon_0}\sqrt{x_3^2+r^2}\bigg|_h^\infty \\ &= -\frac{\sigma}{2\varepsilon_0}\frac{r^2}{x_3+\sqrt{x_3^2+r^2}}\bigg|_h^\infty \\ &= \frac{\sigma}{2\varepsilon_0}\frac{r^2}{h+\sqrt{h^2+r^2}}\end{aligned}$$

2.2.6 一个半径为 r 的圆线框均匀带电, 电荷线密度为 λ, 求线框中心上方距 h 处的电势

$$V=\int_h^\infty \frac{\lambda}{2\varepsilon_0}\frac{x_3 r}{\sqrt{x_3^2+r^2}^3}\,\mathrm{d}x_3=-\frac{\lambda r}{2\varepsilon_0}\frac{1}{\sqrt{x_3^2+r^2}}\bigg|_h^\infty=\frac{\lambda r}{2\varepsilon_0}\frac{1}{\sqrt{h^2+r^2}}$$

2.2.7 一个尖角向下的圆锥体均匀带电, 电荷面密度为 σ, 圆锥高度等于半径为 h, 求 $z=h$ 处的电势

$$\begin{aligned}
V&=\int_{-h}^0 \frac{\lambda(h+x_3)}{2\varepsilon_0}\frac{1}{\sqrt{(h-x_3)^2+(h+x_3)^2}}\,\mathrm{d}x_3\\
&=\int_{-h}^0 \frac{\lambda(h+x_3)}{2\varepsilon_0}\frac{1}{\sqrt{2h^2+2x_3^2}}\,\mathrm{d}x_3\\
&=\frac{\lambda}{2\sqrt{2}\varepsilon_0}\int_{-h}^0 \frac{h}{\sqrt{h^2+x_3^2}}+\frac{x_3}{\sqrt{h^2+x_3^2}}\,\mathrm{d}x_3\\
&=h\ln\left(x_3+\sqrt{h^2+x_3^2}\right)+\sqrt{h^2+x_3^2}\bigg|_{-h}^0\frac{\lambda}{2\sqrt{2}\varepsilon_0}\\
&=\left[h\ln\left(\sqrt{h^2}\right)+\sqrt{h^2}-h\ln\left(-h+\sqrt{2h^2}\right)-\sqrt{2h^2}\right]\frac{\lambda}{2\sqrt{2}\varepsilon_0}\\
&=\left[h\ln\left(\frac{1}{\sqrt{2}-1}\right)+h-\sqrt{2}h\right]\frac{\lambda}{2\sqrt{2}\varepsilon_0}\\
&=\left[1-\sqrt{2}-\ln\left(\sqrt{2}-1\right)\right]\frac{\lambda h}{2\sqrt{2}\varepsilon_0}
\end{aligned}$$

2.2.8 一个半径为 r 的圆柱均匀带电, 电荷体密度为 ρ, 求圆柱中心上方距 h 处的电势

$$\begin{aligned}
V&=\int_{-h}^0 \frac{\rho}{2\varepsilon_0}\frac{r^2}{h+x_3+\sqrt{(h+x_3)^2+r^2}}\,\mathrm{d}x_3\\
&\xlongequal{u=h+x_3}\int_0^h \frac{\rho}{2\varepsilon_0}\frac{r^2}{u+\sqrt{u^2+r^2}}\,\mathrm{d}u\\
&=\int_0^h \frac{\rho r^2}{2\varepsilon_0}\frac{\sqrt{u^2+r^2}-u}{r^2}\,\mathrm{d}u\\
&=\frac{\rho}{2\varepsilon_0}\int_0^h \sqrt{u^2+r^2}-u\,\mathrm{d}u\\
&=u\sqrt{u^2+r^2}+r^2\ln\left(u+\sqrt{u^2+r^2}\right)-u^2\bigg|_0^h\frac{\rho}{4\varepsilon_0}\\
&=\left[h\sqrt{h^2+r^2}+r^2\ln\left(h+\sqrt{h^2+r^2}\right)-h^2-r^2\ln r\right]\frac{\rho}{4\varepsilon_0}
\end{aligned}$$

2.3 静电场的能量

2.3.1 离散电荷体系的静电能

考虑由 N 个点电荷 q_i 构成的静电体系。将电荷从无穷远逐个缓慢搬运到其最终位置, 外力所做的总功即为体系的静电能。

设在放置第 i 个电荷时, 其余电荷已就位, 则该电荷所处位置的当前电势为

$$\varphi_i = \sum_{j=1}^{i-1} \frac{1}{4\pi\varepsilon_0} \frac{q_j}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}.$$

因此, 体系的总静电能为

$$U = \sum_{i=1}^{N} \sum_{j=1}^{i-1} \frac{1}{4\pi\varepsilon_0} \frac{q_i q_j}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}.$$

注意到相互作用能在指标交换下满足 $U_{ij} = U_{ji}$, 而对所有 $i \neq j$ 的求和中每一对指标被计数两次, 故可将上式对称化为

$$U = \frac{1}{2} \sum_{\substack{i,j=1 \\ i \neq j}}^{N} \frac{1}{4\pi\varepsilon_0} \frac{q_i q_j}{|\boldsymbol{r}_i - \boldsymbol{r}_j|} = \frac{1}{2} q_i \varphi_i.$$

2.3.2 连续电荷分布的能量表达式

对离散情形的自然推广给出

$$U = \frac{1}{2} \iiint \rho\varphi \, \mathrm{d}V.$$

2.3.3 场能量

静电势满足泊松方程

$$\nabla^2 \varphi = -\frac{\rho}{\varepsilon_0}.$$

将其代入能量表达式, 得

$$U = -\frac{\varepsilon_0}{2} \iiint \varphi \, \nabla^2 \varphi \, \mathrm{d}V.$$

对右端积分作分部积分。注意到

$$\nabla \cdot (\varphi \nabla \varphi) = (\nabla \varphi)^2 + \varphi \nabla^2 \varphi,$$

于是

$$\varphi \nabla^2 \varphi = \nabla \cdot (\varphi \nabla \varphi) - (\nabla \varphi)^2.$$

代入得

$$U = -\frac{\varepsilon_0}{2} \left[\iiint \nabla \cdot (\varphi \nabla \varphi) \, \mathrm{d}V - \iiint (\nabla \varphi)^2 \, \mathrm{d}V \right].$$

当 $r \to \infty$ 时,

$$\varphi(\boldsymbol{r}) \sim \frac{1}{r}, \qquad \nabla \varphi \sim \frac{1}{r^2}.$$

因此

$$\varphi \nabla \varphi \sim \frac{1}{r^3},$$

对应的无穷远处曲面积分

$$\oiint \varphi \nabla \varphi \cdot \mathrm{d}\boldsymbol{S} \to 0$$

$$U = \frac{\varepsilon_0}{2} \iiint (\nabla \varphi)^2 \, \mathrm{d}V.$$

$$U = \frac{\varepsilon_0}{2} \iiint E^2 \, \mathrm{d}V$$

2.3.4 电场能量密度

这表明：静电能可以视为分布在空间中的电场所携带的能量。

由此自然引入电场的能量密度

$$\boxed{u \equiv \frac{\varepsilon_0}{2}\boldsymbol{E}^2}$$

2.3.5 考虑两个同心球面, 半径分别为 a 和 b, 内球面带有电荷 q , 外球面带有电荷 $-q$, 求总能量

$$U = \frac{\varepsilon_0}{2}\iiint E^2\,\mathrm{d}V = \frac{\varepsilon_0 4\pi}{2}\int_a^b \left(\frac{q}{4\pi\varepsilon_0 r^2}\right)^2 r^2\,\mathrm{d}r = \int_a^b \frac{q^2}{8\pi\varepsilon_0 r^2}\,\mathrm{d}r = \frac{q^2}{8\pi\varepsilon_0}\left(\frac{1}{b}-\frac{1}{a}\right)$$

2.4 导体

2.4.1 导体内部电场为零

若导体内部存在非零电场, 则自由电荷将在电场作用下持续运动, 与静电平衡的假设矛盾。因此, 导体内部必须满足

$$\boldsymbol{E} = -\boldsymbol{\nabla}\varphi = \boldsymbol{0}.$$

2.4.2 导体内部体电荷密度为零

$$\boldsymbol{\nabla}\cdot\boldsymbol{E} = \frac{\rho}{\varepsilon_0},$$

结合 $\boldsymbol{E} = \boldsymbol{0}$,

$$\rho = 0 \qquad (\text{导体内部}).$$

2.4.3 净电荷只能分布在导体表面

既然导体内部体电荷密度为零, 而导体整体可能带有净电荷, 则这些电荷只能分布在导体的表面上。

2.4.4 导体是等势体

由 $\boldsymbol{E} = -\boldsymbol{\nabla}\varphi$ 以及导体内部 $\boldsymbol{E} = \boldsymbol{0}$, 可知导体内部电势处处相同

2.4.5 导体表面外侧电场垂直于表面

考虑导体表面上一点的切向电场分量。若存在非零切向分量, 则自由电荷将在表面沿切向运动, 从而破坏静电平衡。

因此, 导体表面外侧的电场只能沿法向：

$$\boldsymbol{E}_{\parallel} = 0$$

2.4.6 空腔中含点电荷的球形导体

设一不带电的球形导体, 半径为 R, 中心位于原点, 其内部挖去一任意形状的空腔。在空腔内某处放置一点电荷 q。求球外区域的电场分布。

导体处于静电平衡时, 金属内部电场为零, 因而导体整体为等势体。取一紧贴导体内壁、位于金属内部的高斯面, 由高斯定律可得

$$\oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{a} = 0,$$

故内壁所感应的总电荷必为 $-q$。又由于导体整体不带电, 外表面所带电荷总量为 $+q$。

关键在于确定外表面电荷的分布形式。注意到：在所有满足

$$\boldsymbol{E} = 0 \quad (\text{导体内部}), \qquad \varphi = \text{常数} \quad (\text{导体表面})$$

的允许电荷分布中, 实际的静电平衡态对应于体系总静电能的极小值。

球外区域不含自由电荷, 其电势完全由外表面电荷分布决定。若外表面电荷分布破坏球对称性, 则球外电场中将出现非径向分量, 从而在保持总电荷为 q 的约束下增加电场能

$$U = \frac{\varepsilon_0}{2} \int E^2 \, \mathrm{d}V.$$

因此, 能量极小所对应的电荷分布必然保持球对称性。

由此可知, 外表面电荷在球面上均匀分布, 其产生的球外电场与位于球心的点电荷 q 完全相同。因此, 球外任意一点处的电场为

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2} \boldsymbol{e}_r, \qquad r > R.$$

可见, 空腔的形状以及点电荷在腔内的具体位置, 均不影响球外区域的电场分布。

2.4.7 一个半径为 R 的金属球, 带有电荷 q, 这个金属球又被一个厚的同心金属球壳所包围 (球壳内径为 a, 外径为 b)。

(a) 分别求出 R, a, b 球面上的电荷面密度 σ。

(b) 求出球心处的电势, 选无限远处为参考点。

(c) 现在球壳的外表面接地, 电势能为零。(a) 和 (b) 所得结果改变为什么?

(a):

$$\sigma_R = \frac{q}{4\pi R^2}$$
$$\sigma_a = \frac{q}{4\pi a^2}$$
$$\sigma_b = \frac{q}{4\pi b^2}$$

(b):

$$V = \int_b^\infty \frac{q}{4\pi\varepsilon_0 r} \, \mathrm{d}r + \int_R^a \frac{q}{4\pi\varepsilon_0 r} \, \mathrm{d}r = \frac{q}{4\pi\varepsilon_0} \ln\left(\frac{bR}{a}\right)$$

(c):

$$\begin{aligned}\sigma_R &= \frac{q}{4\pi R^2}\\ \sigma_a &= \frac{q}{4\pi a^2}\\ \sigma_b &= 0\\ V &= \frac{q}{4\pi\varepsilon_0}\ln\left(\frac{R}{a}\right)\end{aligned}$$

2.4.8 一个半径为 R 的导体球体, 其内部有两个半径分别为 a 和 b 的圆形空洞, 在 a 空洞的中心放有点电荷 q, 在 b 空洞的中心放有点电荷 q。
(a) 求出电荷面密度 σ_a, σ_b 和 σ_R。
(b) 导体外面的电场是什么?
(c) 每个空洞内的电场是什么?
(d)q_a 和 q_b 受到的力是什么?
(e) 如果让第三个电荷 q 靠近导体, 上面所得结果哪一个会发生变化?

(a):

$$\begin{aligned}\sigma_R &= \frac{2q}{4\pi R^2}\\ \sigma_a &= \frac{q}{4\pi a^2}\\ \sigma_b &= \frac{q}{4\pi b^2}\end{aligned}$$

(b):

$$\boldsymbol{E} = \frac{2q\boldsymbol{r}}{4\pi\varepsilon_0 r^3}$$

(c):

$$\begin{aligned}\boldsymbol{E_a} &= \frac{q\boldsymbol{r}}{4\pi\varepsilon_0 r^3}\\ \boldsymbol{E_b} &= \frac{q\boldsymbol{r}}{4\pi\varepsilon_0 r^3}\end{aligned}$$

(d):0

(e): σ_R, b

2.5 电容

$$C \equiv \frac{Q}{V} \tag{2.6}$$

$$W = \int_0^Q \left(\frac{q}{C}\right)\mathrm{d}q = \frac{Q^2}{2C} \tag{2.7}$$

2.5.1 两个同轴金属管壳, 半径分别为 a 和 b, 求出单位长度的电容。

$$
\begin{aligned}
V &= \int_a^b \frac{Q}{2\varepsilon_0 r}\,\mathrm{d}r \\
&= \frac{Q}{2\varepsilon_0}\ln\left(\frac{b}{a}\right) \\
C &= \frac{Q}{V} \\
&= \frac{2\varepsilon_0}{\ln\left(\frac{b}{a}\right)}
\end{aligned}
$$

2.6 习题

2.6.1 一个均匀带电、边长为 $2a$ 的正方形面, 电荷面密度为 σ。求出距中心高度为 z 处的电场。

$$
\begin{aligned}
&\int_0^a \frac{\sigma}{\pi\varepsilon_0\sqrt{2L^2+z^2}}\boldsymbol{e}_3\frac{2Lz}{L^2+z^2}\,\mathrm{d}L \\
\xlongequal{2L^2=z^2\tan^2\theta} &\int_0^{\arctan\frac{a}{\sqrt{2}z}} \frac{\sigma}{\pi\varepsilon_0\sqrt{z^2\tan^2\theta+z^2}}\boldsymbol{e}_3\frac{2z\tan\theta z}{z^2\tan^2\theta+2z^2}\,\mathrm{d}z\tan\theta \\
=&\int_0^{\arctan\frac{\sqrt{2}a}{z}} \frac{\sigma\cos\theta}{\pi\varepsilon_0 z\cos^2\theta}\boldsymbol{e}_3\frac{2z\tan\theta z}{z^2\tan^2\theta+2z^2}\,\mathrm{d}z\theta \\
=&\int_0^{\arctan\frac{\sqrt{2}a}{z}} \frac{\sigma}{\pi\varepsilon_0\cos\theta}\boldsymbol{e}_3\frac{2\tan\theta}{\tan^2\theta+2}\,\mathrm{d}\theta \\
=&\int_0^{\arctan\frac{\sqrt{2}a}{z}} \frac{2\sigma}{\pi\varepsilon_0}\boldsymbol{e}_3\frac{\tan\theta}{\tan^2\theta\cos\theta+2\cos\theta}\,\mathrm{d}\theta \\
=&\int_0^{\arctan\frac{\sqrt{2}a}{z}} \frac{2\sigma}{\pi\varepsilon_0}\boldsymbol{e}_3\frac{\sin\theta}{\sin^2\theta+2\cos^2\theta}\,\mathrm{d}\theta \\
=&-\int_0^{\arctan\frac{\sqrt{2}a}{z}} \frac{2\sigma}{\pi\varepsilon_0}\boldsymbol{e}_3\frac{1}{1+\cos^2\theta}\,\mathrm{d}\cos\theta \\
=&-\frac{2\sigma}{\pi\varepsilon_0}\boldsymbol{e}_3\arctan\cos\theta\Big|_0^{\arctan\frac{\sqrt{2}a}{z}} \\
=&\frac{2\sigma}{\pi\varepsilon_0}\boldsymbol{e}_3\left(\frac{\pi}{4}-\arctan\frac{z}{\sqrt{2a^2+z^2}}\right)
\end{aligned}
$$

2.6.2 已知电场 $\boldsymbol{E}=\dfrac{A\boldsymbol{e}_r+B\sin\theta\cos\phi\boldsymbol{e}_\phi}{r}$, 求电荷密度

$$
\begin{aligned}
\boldsymbol{\nabla}\cdot\boldsymbol{E}==&\frac{1}{sr}\frac{\partial srE_r}{\partial r}+\frac{1}{s}\frac{\partial E_\phi}{\partial\phi}\\
=&\frac{1}{sr}\frac{\partial sA}{\partial r}+\frac{1}{sr}\frac{\partial B\sin\theta\cos\phi}{\partial\phi}\\
=&\frac{A}{r^2}-\frac{1}{r^2}B\sin\phi
\end{aligned}
$$

2.6.3 一个均匀带电球体, 求出南半球与北半球之间的净相互作用力

$$
\begin{aligned}
E_z=E_r\cos\theta=&\frac{\rho\cos\theta}{4\pi\varepsilon_0r^2}\frac{4\pi r^3}{3}=\frac{\rho\cos\theta r}{3\varepsilon_0}\\
F=&\int_0^R\mathrm{d}r\int_0^{2\pi}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}r^2\sin\theta\rho E_z\,\mathrm{d}\theta\\
=&\int_0^R\mathrm{d}r\int_0^{2\pi}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}r^3\sin\theta\rho\frac{\rho\cos\theta}{3\varepsilon_0}\,\mathrm{d}\theta\\
=&\pi\int_0^R r^3\,\mathrm{d}r\int_0^{\frac{\pi}{2}}\sin\theta\rho^2\frac{2\cos\theta}{3\varepsilon_0}\,\mathrm{d}\theta\\
=&\pi R^4\int_0^{\frac{\pi}{2}}\sin\theta\rho^2\frac{\cos\theta}{6\varepsilon_0}\,\mathrm{d}\theta\\
=&\pi R^4\int_0^{\frac{\pi}{2}}\rho^2\frac{\sin2\theta}{24\varepsilon_0}\,\mathrm{d}2\theta\\
=&-\rho^2R^4\pi\frac{\cos2\theta}{24\varepsilon_0}\bigg|_0^{\frac{\pi}{2}}\\
=&\frac{\rho^2R^4\pi}{12\varepsilon_0}
\end{aligned}
$$

2.6.4 一个半径为 R 的倒置半球面均匀带电, 电荷面密度为 σ。求出北极与球心处的电势差

$$
\begin{aligned}
V_{\text{半球北极}} &= \mathrm{d}\varphi \int_0^{\frac{\pi}{2}} \frac{\sigma R \sin\theta}{2\varepsilon_0} \frac{1}{\sqrt{R^2\left(1-\cos\theta\right)^2 + R^2\sin^2\theta}} R\,\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}} \frac{\sigma \sin^2\theta}{2\varepsilon_0} \frac{1}{\sqrt{\left(1-\cos\theta\right)^2 + \sin^2\theta}} R^2\,\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}} \frac{\sigma \sin\theta}{2\varepsilon_0} \frac{1}{\sqrt{2-2\cos\theta}} R\,\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}} \frac{\sigma 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sqrt{2}\varepsilon_0} \frac{1}{\sqrt{1-\cos^2\frac{\theta}{2}+\sin^2\frac{\theta}{2}}} R\,\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}} \frac{\sigma \sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\sqrt{2}\varepsilon_0} \frac{1}{\sqrt{2\sin^2\frac{\theta}{2}}} R\,\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}} \frac{\sigma \cos\frac{\theta}{2}}{2\varepsilon_0} R\,\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}} \frac{\sigma}{\varepsilon_0} R\,\mathrm{d}\sin\frac{\theta}{2} \\
&= \left.\frac{\sigma \sin\frac{\theta}{2}}{\varepsilon_0} R\right|_0^{\frac{\pi}{2}} \\
&= \frac{\sqrt{2}\sigma}{2\varepsilon_0} R \\
V_{\text{半球球心}} &= \frac{2\pi R^2 \sigma}{4\pi\varepsilon_0 R} = \frac{\sigma R}{2\varepsilon_0} \\
\Delta V &= \frac{\sigma}{2\varepsilon_0}\frac{R^2}{\sqrt{R^2}} + \frac{\sigma R}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0}\frac{R^2}{R+\sqrt{R^2+R^2}} - \frac{\sqrt{2}\sigma}{2\varepsilon_0} R \\
&= \frac{\sigma R}{2\varepsilon_0} - \frac{\sqrt{2}\sigma}{2\varepsilon_0} R \\
&= \frac{\sigma R}{2\varepsilon_0} - \frac{\sqrt{2}\sigma}{2\varepsilon_0} R
\end{aligned}
$$

2.6.5 一个半径为 R 的球体, 电荷密度 $\rho = kr$, 求能量

$$
\begin{aligned}
|\boldsymbol{E}| &= \begin{cases} \dfrac{1}{4\pi\varepsilon_0 x_3^2}\displaystyle\int_0^R 4\pi k r^3\,\mathrm{d}r\,(x_3 \geq R) \\ \dfrac{1}{4\pi\varepsilon_0 x_3^2}\displaystyle\int_0^{x_3} 4\pi k r^3\,\mathrm{d}r\,(x_3 < R) \end{cases} \\
&= \begin{cases} \dfrac{1}{4\varepsilon_0 x_3^2} k R^4\,(x_3 \geq R) \\ \dfrac{1}{4\varepsilon_0 x_3^2} k x_3^4\,(x_3 < R) \end{cases} \\
&= \begin{cases} \dfrac{kR^4}{4\varepsilon_0 x_3^2}\,(x_3 \geq R) \\ \dfrac{k x_3^2}{4\varepsilon_0}\,(x_3 < R) \end{cases} \\
U &= \frac{\varepsilon_0}{2} 4\pi \int_0^R \left(\frac{k x_3^2}{4\varepsilon_0}\right)^2 x_3^2\,\mathrm{d}x_3 + \frac{\varepsilon_0}{2} 4\pi \int_R^\infty \left(\frac{kR^4}{4\varepsilon_0 x_3^2}\right)^2 x_3^2\,\mathrm{d}x_3 \\
&= 2\pi \int_0^R \frac{k^2 x_3^6}{16\varepsilon_0}\,\mathrm{d}x_3 + 2\pi \int_R^\infty \frac{k^2 R^8}{16\varepsilon_0 x_3^2}\,\mathrm{d}x_3 \\
&= \pi \frac{k^2 R^7}{56\varepsilon_0} + \pi \frac{k^2 R^8}{8\varepsilon_0 R} \\
&= \pi \frac{k^2 R^7}{7\varepsilon_0}
\end{aligned}
$$

2.6.6 电势为 $V=\frac{Ae^{-\lambda r}}{r}$, 求电场, 电荷密度, 总电荷

$$
\begin{aligned}
\boldsymbol{E} &= -\boldsymbol{\nabla} V \\
&= \frac{\partial \dfrac{Ae^{-\lambda r}}{r}}{\partial r}\boldsymbol{e}_r \\
&= \frac{\lambda r A e^{-\lambda r} + Ae^{-\lambda r}}{r^2}\boldsymbol{e}_r \\
\rho &= \varepsilon_0 \boldsymbol{\nabla}^2 V \\
&= \varepsilon_0 \frac{1}{r^2}\frac{\partial r^2 a_r}{\partial r} \\
&= \varepsilon_0 \frac{1}{r^2}\frac{\partial \lambda r A e^{-\lambda r} + Ae^{-\lambda r}}{\partial r} \\
&= \varepsilon_0 \frac{\lambda A e^{-\lambda r} - \lambda^2 r A e^{-\lambda r} - \lambda A e^{-\lambda r}}{r^2} \\
&= -\frac{\varepsilon_0 \lambda^2 A e^{-\lambda r}}{r}\ (r \neq 0) \\
\rho &= \frac{\varepsilon_0 A 4\pi \delta^{(3)}(r)}{r} - \frac{\varepsilon_0 \lambda^2 A e^{-\lambda r}}{r} \\
Q &= 4\pi \int_0^\infty \rho r^2 \,\mathrm{d}r \\
&= 0
\end{aligned}
$$

2.6.7 两条平行于 z 轴的无限长均匀带电线, 电荷线密度分别为 $+\lambda$ 和 $-\lambda$, 距离为 $2d$。
(a) 求出任意一点的电势。
(b) 证明等势面为圆柱面, 对给定的电势 V, 给出圆柱面的半径和轴的位置。

(a):

$$
\begin{aligned}
V_{+\lambda} &= -\frac{\lambda}{2\pi\varepsilon_0}\ln|\boldsymbol{r} - d\boldsymbol{e}_1| \\
V_{-\lambda} &= \frac{\lambda}{2\pi\varepsilon_0}\ln|\boldsymbol{r} + d\boldsymbol{e}_1| \\
V &= -\frac{\lambda}{2\pi\varepsilon_0}\ln|\boldsymbol{r} - d\boldsymbol{e}_1| + \frac{\lambda}{2\pi\varepsilon_0}\ln|\boldsymbol{r} + d\boldsymbol{e}_1| \\
&= \frac{\lambda}{4\pi\varepsilon_0}\ln\left[\frac{(x_1+d)^2 + x_2^2}{(x_1-d)^2 + x_2^2}\right]
\end{aligned}
$$

(b):

$$
\begin{aligned}
(x_1+d)^2+x_2^2 &= k\left[(x_1-d)^2+x_2^2\right] \\
x_1^2+2x_1d+d^2+x_2^2 &= kx_1^2-2kx_1d+kd^2+kx_2^2 \\
0 &= (k-1)x_1^2-2(k+1)x_1d+(k-1)d^2+(k-1)x_2^2 \\
0 &\xlongequal{u=\frac{k+1}{k-1}} x_1^2-2ux_1d+d^2+x_2^2 \\
u^2-d^2 &= (x_1-u)^2+x_2^2 \\
R &= \sqrt{u^2-d^2} \\
&= \sqrt{\left(\frac{k+1}{k-1}\right)^2-d^2} \\
&= \sqrt{\left(\frac{e^{\frac{4\pi\varepsilon_0 V}{\lambda}}+1}{e^{\frac{4\pi\varepsilon_0 V}{\lambda}}-1}\right)^2-d^2}
\end{aligned}
$$